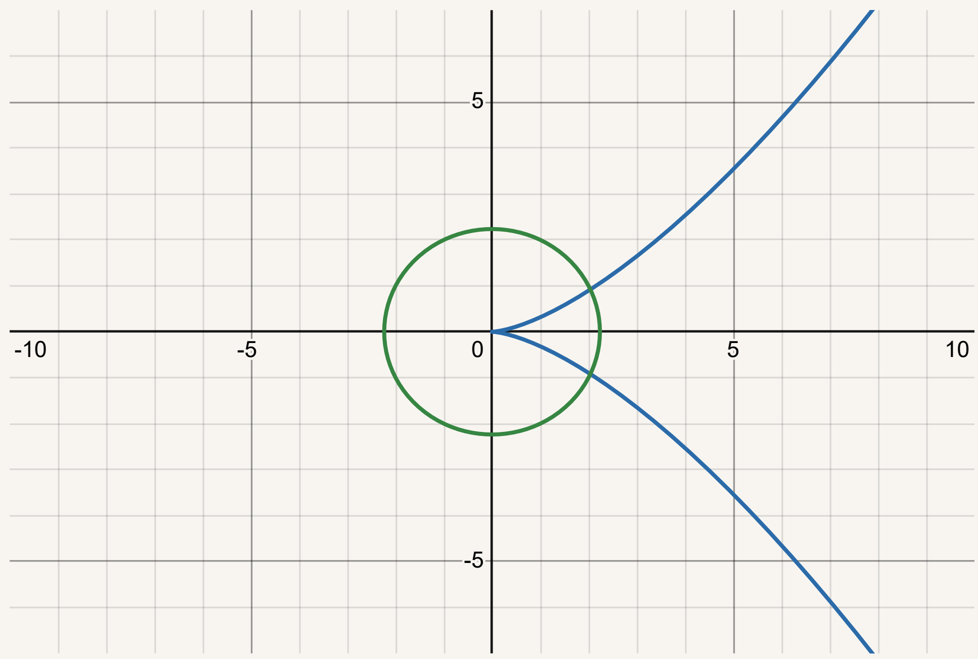
2502 geometry and analysis.N

Spring and summer '25:



> Short link: https://dub.sh/spring25
> Typeset by [Rupadarshi Ray]()
> while throwing ideas with my peers [Manan Jain](), Pahul Arora and Naman Narang
> at [IISER Mohali]()
> Last updated: 10:38 AM - June 27, 2025

timeline

'25.

The spring semester in IISER Mohali runs from January to early May:

ended 51 days ago

Then we have summer:

Summer holidarys will start from Wed 7 May, '25 and end on Mon 4 Aug, '25.

38 days left in summer holidays of 90 days

57.77% complete

And then the *thesis year begins* (for 3/4 of us above, anyways):

MS thesis period will start from Mon 4 Aug, '25 and end on Sat 4 Apr, '26 tentatively.

38 days to go

FYI: It is a 244 days of MS thesis year.

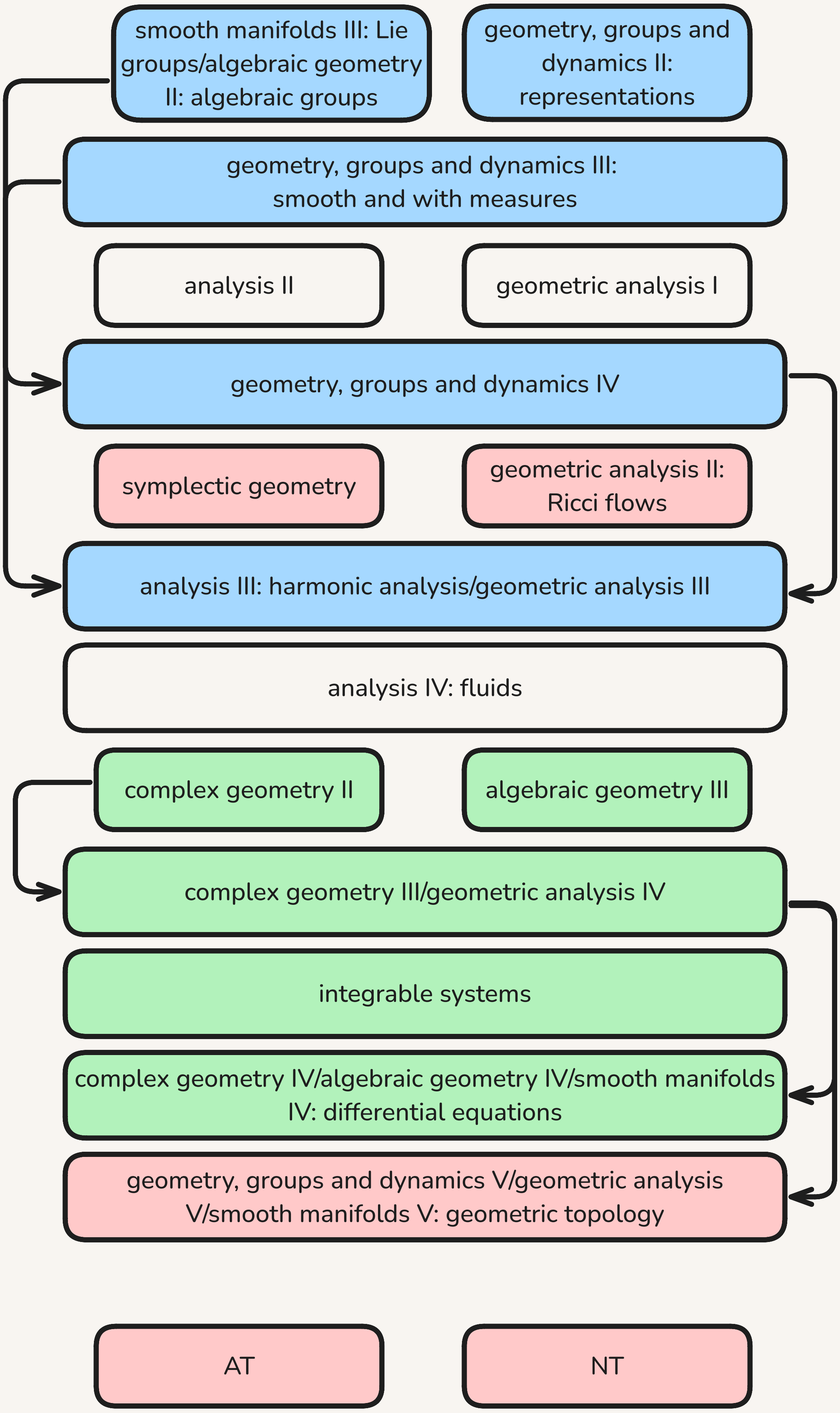
notebooks

This piece of notebook is a part of a *bigger* set of notebooks

notebooks

- *red* (manifolds)
- *complex geometry I: complex 1-manifolds* in the short link https://dub.sh/riemann
- *blue* (GGD)
- *green* (CG/AG, integrable systems)

with topics:



⚠ **Warning**

Every note in what follows is ***complete*** and ***correct*** *asymptotically*...

contents: complex geometry I: complex 1-manifolds

AKA *Riemann surfaces*

- $\sum_{n\geq 0} z^{n!}$
- Universal cover of punctured plane
- The usual definition of Riemann surfaces
  - Holomorphic functions on a complex 1-manifold
  - Differential forms on complex 1-manifolds
  - Cohomology groups of Riemann surfaces
- Concept of a Riemann surface: usual definition vs as a ringed space
- Classification of Riemann surfaces
- Constructing meromorphic functions on Riemann surfaces
- periods of elliptic curves
- book readings
  - Simon Donaldson - Riemann surfaces

The complex tangent space on $\mathbb{C}$

Definition. The complex tangent and cotangent space on $\mathbb{C}$

On the complex tangent space at $z \in \mathbb{C}$

$$T_z\mathbb{R}^2 \otimes \mathbb{C}$$

we define the basis

$$\begin{aligned}\frac{\partial}{\partial z} &:= \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)\\ \frac{\partial}{\partial \bar{z}} &:= \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)\end{aligned}$$

And we think of them as operators on differentiable functions $\mathbb{C} \to \mathbb{C}$. We have

$$\begin{aligned}\mathrm{d}z &= \mathrm{d}x + i\mathrm{d}y\\ \mathrm{d}\bar{z} &= \mathrm{d}x - i\mathrm{d}y\end{aligned}$$

as basis for $T_z^*\mathbb{R}^2 \otimes \mathbb{C}$.

But where is this coming from? Well, we slowly progress:

Let $U \subseteq \mathbb{R}^2$ be open.

- *for functions to $\mathbb{R}^2$*: $\begin{bmatrix} f_1 \\ f_2 \end{bmatrix} \in \mathcal{C}^1(\mathbb{R}^2, \mathbb{R}^2)$
  - we have the total derivative $\mathrm{d}\begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_2}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix}$
- for $f_1 + if_2 \in \mathcal{C}^1(\mathbb{R}^2, \mathbb{R}) \otimes \mathbb{C} \cong \mathcal{C}^1(\mathbb{R}^2, \mathbb{R}) + i\mathcal{C}^1(\mathbb{R}^2, \mathbb{R})$
  - we have the derivative

    $$\mathrm{d}(f_1 + if_2) = \left(\frac{\partial f_1}{\partial x} + i\frac{\partial f_2}{\partial x}\right)\mathrm{d}x + \left(\frac{\partial f_1}{\partial y} + i\frac{\partial f_2}{\partial y}\right)\mathrm{d}y$$

    which is using the identification of $T_p^*\mathbb{R}^2 \otimes \mathbb{C} \cong \mathbb{R}\{\mathrm{d}x, \mathrm{d}y\} \otimes \mathbb{C} \cong \mathbb{C}\{\mathrm{d}x, \mathrm{d}y\}$
  - this gives four $\mathbb{R}$-linear maps

    $$\frac{\partial(-)_1}{\partial x} : \mathcal{C}^1(\mathbb{R}^2, \mathbb{R}) \otimes \mathbb{C} \to \mathbb{R}$$

    $\mathbb{R}$-span the complexified tangent space $T_p\mathbb{R}^2 \otimes \mathbb{C}$
- for $f \in \mathcal{C}^1(\mathbb{R}^2, \mathbb{C})$ where we can use the field multiplication of $\mathbb{C}$!
  - $$\begin{aligned}\mathrm{d}f &= \frac{\partial f}{\partial x}\mathrm{d}x + \frac{\partial f}{\partial y}\mathrm{d}y\\ &= \frac{1}{2}\left(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\right)(\mathrm{d}x + i\mathrm{d}y) + \frac{1}{2}\left(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\right)(\mathrm{d}x - i\mathrm{d}y)\\ &=: \frac{\partial f}{\partial z}\mathrm{d}z + \frac{\partial f}{\partial \bar{z}}\mathrm{d}\bar{z}\end{aligned}$$

    which is an element of $T_p\mathbb{R}^2 \otimes \mathbb{C} \cong \mathbb{C}\{\mathrm{d}x, \mathrm{d}y\}$
  - there are $\mathbb{C}$-linear maps

    $$\frac{\partial(-)}{\partial x}, \frac{\partial(-)}{\partial y} : \mathcal{C}^1(\mathbb{R}^2, \mathbb{R}) \otimes \mathbb{C} \to \mathbb{C}$$

and another pair of $\mathbb{C}$-linear maps

$$\frac{\partial(-)}{\partial z}, \frac{\partial(-)}{\partial \bar{z}} : \mathcal{C}^1(\mathbb{R}^2, \mathbb{R}) \otimes \mathbb{C} \to \mathbb{C}$$

both pairs separately $\mathbb{C}$-span the complexified tangent space $T_p\mathbb{R}^2 \otimes \mathbb{C}$

Complex differential forms on $\mathbb{C}$

Complex differential forms

$$f\mathrm{d}z + g\mathrm{d}\bar{z}$$

mimic the differential forms on $\mathbb{R}^2$, while holomorphic differential forms

$$f\mathrm{d}z$$

mimic the forms on $\mathbb{R}$.

The differential complex of smooth complex valued functions looks like this

$$\begin{array}{ccccc}
\Omega^0(U,\mathbb{C}) & \to & \Omega^{1,0}(U,\mathbb{C}) \oplus \Omega^{0,1}(U,\mathbb{C}) & \to & \Omega^2(U,\mathbb{C}) \\
\phi & \overset{\mathrm{d}}{\mapsto} & \mathrm{d}\phi = \frac{\partial \phi}{\partial z}\mathrm{d}z + \frac{\partial \phi}{\partial \bar{z}}\mathrm{d}\bar{z} & & \\
 & & f\mathrm{d}z + g\mathrm{d}\bar{z} & \overset{\mathrm{d}}{\mapsto} & \left(\frac{\partial g}{\partial z} - \frac{\partial f}{\partial \bar{z}}\right)\mathrm{d}z \wedge \mathrm{d}\bar{z} \\
 & & & & h\mathrm{d}z \wedge \mathrm{d}\bar{z}
\end{array}$$

Integration on chains, differentiation and boundary of chains mixes up in the statement

(Green-Stroke's theorem for integration of complex forms on chains)

$$\int_M \mathrm{d}\omega = \int_{\partial M} \omega$$

(**Cauchy–Pompeiu formula**) For $f \in \mathcal{C}^1(D)$ on a closed disk $D \subseteq \mathbb{C}$ we have

$$f(u) = \frac{1}{2\pi i}\int_{\partial D} \frac{f(z)}{z-u}\mathrm{d}z - \frac{1}{\pi}\int_D \frac{\partial f}{\partial \bar{z}} \frac{1}{z-u}\mathrm{d}x \wedge \mathrm{d}y$$

which is just Stroke's theorem/Green's used for

$$\mathrm{d}\left(\frac{f(z)}{z-u}\mathrm{d}z\right) = \left(f(z)\frac{\partial}{\partial \bar{z}}\frac{1}{z-u} + \frac{\partial f}{\partial \bar{z}}\frac{1}{z-u}\right)\mathrm{d}z \wedge \mathrm{d}\bar{z}$$

the statement that the function

$$\frac{\partial}{\partial \bar{z}}\frac{1}{z-u} = \pi\delta_{z-u}$$

is the Dirac delta.

Hodge dual

Definition.

$$\star(p\mathrm{d}x + q\mathrm{d}y) := -q\mathrm{d}x + p\mathrm{d}y$$

or equivalently

$$\star\mathrm{d}z = -i\mathrm{d}z$$
$$\mathrm{d}\bar{z} = i\mathrm{d}\bar{z}$$

A complex 1-form ω satisfies

$$\mathrm{d}\omega = 0$$
$$\star\omega = -i\omega$$

if and only if it looks like

$$\omega = f\mathrm{d}z$$

where f is a holomorphic function.

Complex conjugate

$$\overline{f\mathrm{d}x + g\mathrm{d}y} = \bar{f}\mathrm{d}x + \bar{g}\mathrm{d}y$$

so this means

$$\begin{aligned}\overline{\mathrm{d}z} &= \overline{\mathrm{d}x + i\mathrm{d}y} = \mathrm{d}x - i\mathrm{d}y = \mathrm{d}\bar{z}\\ \overline{\mathrm{d}\bar{z}} &= \overline{\mathrm{d}x - i\mathrm{d}y} = \mathrm{d}x + i\mathrm{d}y = \mathrm{d}z\end{aligned}$$

and

$$\overline{f\mathrm{d}z + g\mathrm{d}\bar{z}} = \bar{f}\mathrm{d}\bar{z} + \bar{g}\mathrm{d}z$$

Integration also plays nice?

$$\int\bar{\omega} = \overline{\int\omega}$$

Integral of differential forms on $\mathbb{C}$

Definition. **$\mathcal{C}^1$-singular homology on $U \subseteq \mathbb{C}$**

Let R be any of the rings $\mathbb{Z}, \mathbb{R}$ or $\mathbb{C}$.

- The **0-chains** on U

$$\mathrm{sing}_0(U, R)$$

  are R-linear combinations of points of U.
- The **1-chains** on U

$$\mathrm{sing}_1(U, R)$$

  as a formal R-linear combination of piecewise $\mathcal{C}^1$-curves $[0,1] \to U$.
- The **2-chains** on U

$$\mathrm{sing}_2(U, R)$$

  are R-linear combinations of piecewise $\mathcal{C}^1$ maps from $[0,1]^2$ to U
- We define

$$\mathrm{sing}_2(U, R) \xrightarrow{\partial} \mathrm{sing}_1(U, R) \xrightarrow{\partial} \mathrm{sing}_0(U, R)$$

Definition. **Integration of k-forms on $\mathcal{C}^1$-singular chains on $U \subseteq \mathbb{C}$**

- Integration of a *0-form* on a 0-chain are just sums of values
- Integration of a *1-form* $f\mathrm{d}z + g\mathrm{d}\bar{z}$ on a 1-chain is defined using integration on curves and extended R-linearly

$$\int_{\sum_{i=1}^n \alpha_i \gamma_i} f\mathrm{d}z + g\mathrm{d}\bar{z} = \sum_{i=1}^n \alpha_i \int_{\gamma_i} f\mathrm{d}z + g\mathrm{d}\bar{z}$$

- We may integrate *2-form* on a 2-chain B by

$$\int_B f\mathrm{d}z \wedge \mathrm{d}\bar{z}$$

From

$$\mathrm{d}z = \mathrm{d}x + i\mathrm{d}y$$
$$\mathrm{d}\bar{z} = \mathrm{d}x - i\mathrm{d}y$$
$$\mathrm{d}z \wedge \mathrm{d}\bar{z} = -2i\mathrm{d}x \wedge \mathrm{d}y$$

we have

$$\int_B f\mathrm{d}z \wedge \mathrm{d}\bar{z} = -2i \int_B f\mathrm{d}x \wedge \mathrm{d}y$$

Holomorphic functions on $\mathbb{C}$

Definition. Holomorphic functions on $\mathbb{C}$

Given an open subset $U \subseteq \mathbb{C}$, a function

$$f : U \to \mathbb{C}$$

is called **holomorphic** if it satisfies any of the equivalent conditions

- $$\lim_{h \to 0, h \in \mathbb{C} \setminus \{0\}} \frac{f(z+h) - f(z)}{h}$$

  exists for all point $z \in U$
- $\iff$ f is *analytic* that is $\forall z_0 \in U, \exists \epsilon > 0$ such that

  $$f(z) = \sum_{n \geq 0} a_n (z - z_0)^n$$

  for all $z \in B(\epsilon, z)$
- $\iff$ f satisfies the Cauchy-Riemann equations

  $$\frac{\partial}{\partial \bar{z}} f = 0$$

- $\iff$ it satisfies Cauchy integral formula, that is, $f \in C^1(U)$, for all $z_0 \in U$, $\exists \epsilon > 0$

  $$f(z_0) = \frac{1}{2\pi i} \int_{\partial B(\epsilon, z_0)} \frac{f(z)}{z - z_0}$$

- $\iff$ the complexified total derivative (whose Jacobian is) $\mathfrak{D}^{\mathbb{C}}_{z_0} f : T_{z_0}\mathbb{R}^2 \otimes \mathbb{C} \to T_{f(z_0)}\mathbb{R}^2 \otimes \mathbb{C}$

  $$\begin{bmatrix} \frac{\partial}{\partial z} f & \frac{\partial}{\partial \bar{z}} f \\ \underbrace{\frac{\partial}{\partial z} \bar{f}}_{\overline{\frac{\partial}{\partial \bar{z}} f}} & \underbrace{\frac{\partial}{\partial \bar{z}} \bar{f}}_{\overline{\frac{\partial}{\partial z} f}} \end{bmatrix}$$

  is diagonal $\iff$ it is a $\mathbb{C}$-linear map and then we have the complex total derivative (whose Jacobian is)

  $$\begin{bmatrix} \frac{\partial}{\partial z} f \end{bmatrix}$$

object	some derivative operators	equivalent to Cauchy-Riemann operator	Cauchy-Riemann equations is when the Cauchy-Riemann operator gives 0
complex function $f = u + iv$	The *total derivative* $\mathfrak{D} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} u_x & u_y \\ v_x & v_y \end{bmatrix}$ which splits into $\frac{\partial}{\partial z}, \frac{\partial}{\partial \bar{z}}$	$\frac{\partial}{\partial \bar{z}} f = \frac{1}{2}\Big((u_x - v_y) + i($	Holomorphic

object	some derivative operators	equivalent to Cauchy-Riemann operator	Cauchy-Riemann equations is when the Cauchy-Riemann operator gives 0
a (complex) differential $(1,0)$-form $$f(z)\mathrm{d}z$$		$$\mathrm{d}(f(z)\mathrm{d}z) = -\frac{\partial f}{\partial \bar{z}}\mathrm{d}z \wedge \mathrm{d}\bar{z}$$	closed differential form
(real) differential 1-form $$\omega = u\mathrm{d}x - v\mathrm{d}y$$ and its Hodge dual $$\star\omega = v\mathrm{d}x + u\mathrm{d}y$$		$$\mathrm{d}\omega = (u_y + v_x)\mathrm{d}x \wedge \mathrm{d}y$$ $$\mathrm{d}\star\omega = (v_y - u_x)\mathrm{d}x \wedge \mathrm{d}y$$	closed, coclosed differential form
(real) vector field $$\bar{f} = \begin{bmatrix} u \\ -v \end{bmatrix}$$		$$\mathrm{curl}\bar{f} = -v_x - u_y$$ $$\mathrm{div}\bar{f} = u_x - v_y$$	*incompressible* (area preserving/symplectic), *solenoid* vector field
two real functions $$u(x,y), v(x,y)$$	$$\Delta u = u_{xx} + u_{yy}$$ $$\Delta v = v_{xx} + v_{yy}$$ $$\mathrm{grad}u \cdot \mathrm{grad}v = u_x v_x + u$$	$$\mathrm{grad}u + \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\mathrm{grad}v =$$	gradient of u rotated 90 degrees is the gradient of v - both are harmonic and conjugate to each other
one real function $$u(x,y)$$			orthogonal real (harmonic?) functions $$\|\mathrm{grad}u\|_{z_0} = \|\mathrm{grad}v\|_{z_0}$$ at some $z_0 \in \mathbb{C}$

Holomorphic mapping onto disk

Riemann mapping onto disk

(Riemann mapping theorem) Let $\Omega \subset \mathbb{C}$ be a simply connected open, proper subset of $\mathbb{C}$. Then there exists a biholomorphism from Ω onto the unit disk

$$f : \Omega \to D$$

Moreover, given $a \in \Omega$ there is a unique biholomorphism

$$\begin{aligned} f : \Omega \to D \\ f(a) = 0, f'(a) > 0 \end{aligned}$$

(uniqueness) For a give $a \in \Omega$, let there are two biholomorphisms

$$f, g : \Omega \to D$$

then

$$f \circ g^{-1} : D \to D$$

is a automorphism of the disk such that

$$\begin{aligned} f \circ g^{-1}(0) = f(a) = 0 \\ (f \circ g^{-1})'(0) = f'(g^{-1}(0)) \frac{1}{g'(g^{-1}(0))} < 1 \end{aligned}$$

- By

  (Swartz inequality) Any holomorphic map

  $$f : D \to D, f(0) = 0$$

  satisfies both

  $$\begin{aligned} |f'(0)| \leq 1 \\ |f(z)| \leq |z| \end{aligned}$$

  Moreover, equality in *either* of these inequalities $\iff$ its a rotation $f(z) = cz, |c| = 1$.

  we conclude

  $$f \circ g^{-1}(w) = cw, |c| = 1$$

- This means

  $$f(z) = cg(z)$$

  but

  $$0 < f'(a) = cg'(a), g'(a) > 0 \implies c = 1$$

  this $f = g$.

[1]

constructing biholomorphisms into the unit disk

- Assume $0 \notin \Omega$. Then there exists a branch of square root $\sqrt{}$ on Ω. Let the image be $\sqrt{\Omega}$.

- The square root on Ω is injective.
- If $w \in \sqrt{\Omega}$ then $-w \notin \sqrt{\Omega}$, as otherwise there are $z_1, z_2 \in \Omega$ such that

  $$\sqrt{z_1} = w = -\sqrt{z_2} \implies z_1 = z_2 \text{ or } w = -w$$

  which implies $0 \in \Omega$ which is a contradiction.
- As $\sqrt{\Omega}$ is open

  $$\sqrt{\Omega} \cap B_\delta(w_0) = \emptyset$$

- So

  $$\begin{gathered} \left|\sqrt{z} - w_0\right| > \delta \\ \frac{\delta}{\left|\sqrt{z} - w_0\right|} < 1 \end{gathered}$$

- Thus

  $$\begin{aligned} f : \Omega &\to D \\ z &\mapsto \frac{\delta}{\sqrt{z} - w_0} \end{aligned}$$

  is injective into the unit disk D.

by maximizing modulus

Proposition: The unique biholomorphism in

> **(Riemann mapping theorem) Let $\Omega \subset \mathbb{C}$ be a simply connected open, proper subset of $\mathbb{C}$. Then there exists a biholomorphism from Ω onto the unit disk**
>
> $$f : \Omega \to D$$
>
> **Moreover, given $a \in \Omega$ there is a unique biholomorphism**
>
> $$\begin{gathered} f : \Omega \to D \\ f(a) = 0, f'(a) > 0 \end{gathered}$$

is the one attains the **maximum** of

$$\begin{aligned} \{h : \Omega \to D \mid h \text{ is holomorphic, injective }\} &\to \mathbb{R} \\ h &\mapsto |h'(a)| \end{aligned}$$

(that is this maximum is also surjective).

- *(surjectivity)* For such an Ω, let

  $$f : \Omega \to D$$

  be an *injective* holomorphic map such that

  $$f'(0) = \sup_{g \in \mathcal{F}} |g'(0)|$$

  - Assume $\alpha \in D \setminus f(\Omega)$. Let $R_\alpha : D \to D$ be an holomorphic automorphism such that $R_\alpha(\alpha) = 0$, then

    $$R_\alpha(f(\Omega))$$

    avoids the origin $0 \in D$.
  - As Ω is simply connected

Montel's normal families of holomorphic functions and a converging sequence in them

Definition. Montel's normal family of holomorphic functions

A family of holomorphic functions on a open $\Omega \subseteq \mathbb{C}$

$$\mathcal{F} = \{f_\alpha \mid \alpha \in \mathcal{A}\} \subset \mathscr{O}(\Omega)$$

is a **normal family** if for each compact $K \subset \Omega$, the family is *uniformly bounded on K*

$$\sup_{z \in K} \sup_{\alpha \in \mathcal{A}} |f_\alpha(z)| < \infty$$

(Montel's normal family theorem) For any normal family of holomorphic functions

Definition. Montel's normal family of holomorphic functions

A family of holomorphic functions on a open $\Omega \subseteq \mathbb{C}$

$$\mathcal{F} = \{f_\alpha \mid \alpha \in \mathcal{A}\} \subset \mathscr{O}(\Omega)$$

is a normal family if for each compact $K \subset \Omega$, the family is *uniformly bounded on K*

$$\sup_{z \in K} \sup_{\alpha \in \mathcal{A}} |f_\alpha(z)| < \infty$$

there exists a sequence f_n in $\mathcal{F}$ that converges uniformly on compact subsets of Ω.

Let f be holomorphic in a disk around $z \in \mathbb{C}$ then

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_{S_r^1} \frac{f(w)}{(w-z)^{n+1}} \mathrm{d}w$$

- So $\{f_\alpha^{(k)} \mid \alpha \in \mathcal{A}\}$ is uniformly bounded on compact subsets of Ω for all $k \geq 0$.
-

using a normal family

- We assume $\Omega \subset D$ and $0 \in \Omega$. Consider the family

$$\mathcal{F} := \{f : \Omega \to D \text{ holomorphic, injective} \mid f(0) = 0, f'(0) > 0\}$$

  - In particular, $f'(0) \in \mathbb{R}$. As $\mathsf{Id}_\Omega \in \mathcal{F}$ it is not empty.
- As

$$|f(z)| < 1 \implies \sup_{z \in D} \sup_{f \in \mathcal{F}} |f(z)| < 1$$

so $\mathcal{F}$ is a normal family

Definition. Montel's normal family of holomorphic functions

A family of holomorphic functions on a open $\Omega \subseteq \mathbb{C}$

$$\mathcal{F} = \{f_\alpha \mid \alpha \in \mathcal{A}\} \subset \mathscr{O}(\Omega)$$

is a normal family if for each compact $K \subset \Omega$, the family is *uniformly bounded on K*

$$\sup_{z \in K} \sup_{\alpha \in \mathcal{A}} |f_\alpha(z)| < \infty$$

- Consider the sequence

$$f_m \in \mathcal{F}$$

such that

$$f'_m(0) \xrightarrow{m \to \infty} \sup_{h \in \mathcal{F}} f'(0)$$

This sequence is also a normal family

- By

(Montel's normal family theorem) For any normal family of holomorphic functions

Definition. **Montel's normal family of holomorphic functions**

A family of holomorphic functions on a open $\Omega \subseteq \mathbb{C}$

$$\mathcal{F} = \{f_\alpha \mid \alpha \in \mathcal{A}\} \subset \mathscr{O}(\Omega)$$

is a normal family if for each compact $K \subset \Omega$, the family is *uniformly bounded on K*

$$\sup_{z \in K} \sup_{\alpha \in \mathcal{A}} |f_\alpha(z)| < \infty$$

there exists a sequence f_n in $\mathcal{F}$ that converges uniformly on compact subsets of Ω.

there is a subsequence f_n which converges

$$f_n \xrightarrow{n \to \infty} f_\infty \in \mathscr{O}(\Omega)$$

uniformly on compact subsets of Ω, such that

$$f'_\infty(0) = \sup_{f \in \mathcal{F}} f'(0)$$

- Then $f_\infty(0) = 0, f'_\infty(0) > 0$, hence the map is not constant.
- As $f_\infty(\Omega) \subset \bar{D}$ by

(Open mapping theorem for open subsets in $\mathbb{C}$) Let f be *non-constant* holomorphic function on a connected open set U then $f(U)$ is open (and connected).

$f(\Omega) \subseteq D$, thus we have a holomorphic

$$f : \Omega \to D$$

-

by solving a Dirichlet problem

[2]

- Let $\Omega \subset D$ and $0 \in \Omega$. We solve the Dirichilet problem for

$$\log|\zeta| \text{ for } \zeta \in \partial\Omega$$

and obtain a harmonic

$$u : \Omega \to \mathbb{R}$$

such that

$$u(z) \xrightarrow{z \to \zeta \in \partial\Omega} \log|\zeta|$$

  - Since Ω is simply connected there is a harmonic conjugate v to u, making

$$f := z\exp(-(u+iv)) : \Omega \to \mathbb{C}$$

    holomorphic on Ω such that

$$|f(z)| \xrightarrow{z \to \zeta \in \partial\Omega} 1$$

  - Hence, $f(\Omega) \subset \bar{D}$ and by open mapping or maximum modulus $f(\Omega) \subset D$. Thus

$$f : \Omega \to D$$

  - Applying the argument principle to ...

---

1. https://www.math.stonybrook.edu/~bishop/classes/math626.F08/rmt.pdf ↩
2. complex analysis - Solution verification —— Proof of the Riemann mapping theorem using Perron's solution of the Dirichlet problem. - Mathematics Stack Exchange ↩

Summer Seminar on Riemann surfaces

Seminar website: https://rupadarshiray.github.io/academia/summer-25-seminar

summer-25-seminar

A seminar on *Riemann surfaces* by Prof Kapil Hari is organized at IISER Mohali during the summer of 2025. Details follow.

(Possible) pre-requisites:

- Open mapping theorem in complex analysis
- Definition of Riemann surfaces and understanding that one smooth 2-manifold may have multiple Riemann surface structures (such as $\mathbb{C}$ is not biholomorphic to D, *different* complex tori, etc)
- History of Jacobi and Abelian integrals

notes

The notes for the topics discussed will be kept here: https://dub.sh/riemann

plan

- *June to July*: Essentially some topics from chapters 4 to 15 from *Narasimhan - Compact Riemann Surfaces*
  - sheaves
  - (Etale space) definition of Riemann surface of a global holomorphic function (in *Ahlfors*)
  - Riemann surfaces of algebraic functions
  - Riemann-Roch theorem
  - The Jacobian of a Riemann surface and Abel's theorem
  - Torelli theorem - Wikipedia

  August to November: The previous plan was to follow *Ramanan's global analysis* but we shall consider the interests of the participants.

references

Possible references for topics in Riemann surfaces:

- https://www.mat.univie.ac.at/~armin/lect/advanced_complex_analysis.pdf
- Books on Riemann surfaces by (Narasimhan may be too daunting)
  - Forster
  - Miranda
  - Wilhelm Schlag
  - https://mathweb.tifr.res.in/~srinivas/rsfull.pdf (uses ringed spaces definition of Riemann surfaces where the rings are rings of functions on it)
  - https://math.berkeley.edu/~teleman/math/Riemann.pdf
  - Donaldson
  - Jost
  - Renzo Cavalieri, Eric Miles (Hurwitz theory)
  - ...a hundred more books and notes available online
- These lectures assume covering space theory (algebraic topology) and uniformization theory and does (pre-moduli space) classification of Riemann surfaces: https://www.youtube.com/playlist?list=PLbMVogVj5nJSm4256vuITlsovUT1xVkUL

- Elliptic Functions and Integrals, Uniformization Theorem etc: https://www.bilibili.com/video/BV1fW41197nr/?spm_id_from=333.337.search-card.all.click

more events to attend in summer

- Summer School on Rigidity of Discrete Groups, June 30 – July 4, 2025 at IISER Mohali

epilogue

- A message from Kapil Hari, the Dean academics: https://iisermmag.wordpress.com/2012/08/23/message-from-the-dean-academics/

Lectures and discussions

- Lecture 1 (18 June)
  - Motivation: Analytic continuation and maximal domain of holomorphic functions,
  - Global holomorphic functions on the Etale space $\mathscr{O}$
- Lecture 2 (23 June)
  - The Etale spaces $\mathscr{M}, \overline{\mathscr{M}}$ where the latter solves the problem of finding the maximal domain of a holomorphic function
  - Concept of a Riemann surface as a ringed space *locally* modelled on $(D, \mathscr{O})$
- Lecture 3 (25 June)
  - Meromorphic functions on Riemann surfaces
- Discussion 1 (28 June)
  - Riemann surfaces of $\log, \sqrt{z}, \sqrt{z^2-1}, \sqrt{z(z^2-1)}$ and branching of holomorphic functions
- Discussion 2 (5 July)
  - ?
- Discussion $\#$
  - Cohomology groups of Riemann surfaces

Analytic continuation and *maximal* domain of holomorphic functions

Example

$$\frac{1}{z}$$

for disk around $z = 1$, $z = 2$

Question

Given some complex analytic functions make sense outside the *disk of convergence*, what is the maximal domain of definition of an *holomorphic function*?

Riemann's answer was that it is a *Riemann surface* or a complex 1-manifold in modern language.

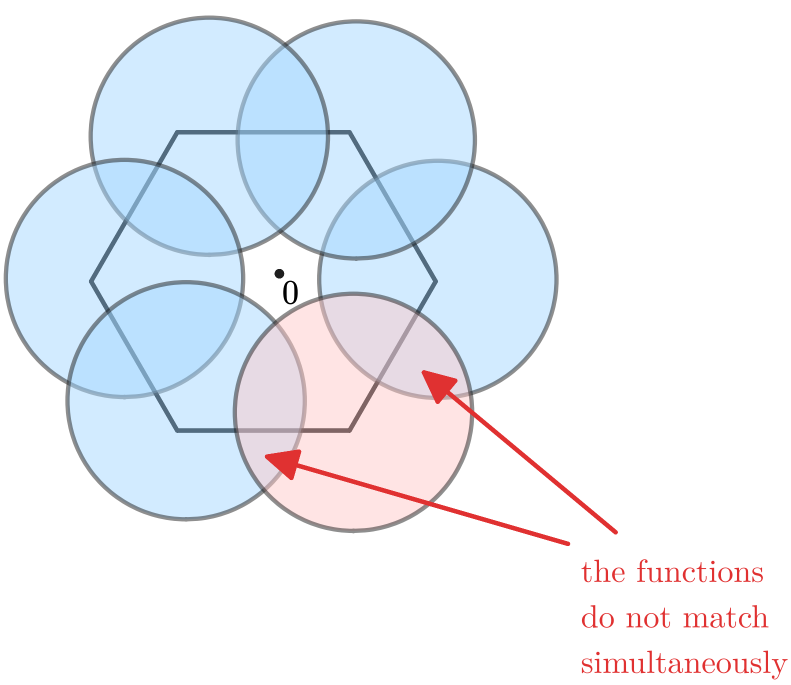
Example

$$\log z := \int_{1,z} \frac{1}{z}$$

and $\log z$ is actually defined on any disk with 0 in its boundary
And it may look as *if* it is defined on $\mathbb{C} \setminus \{0\}$... but that is not true! Because

$$\int_{S^1} \frac{1}{z} = \frac{1}{2\pi i} \neq 0$$

What goes wrong to continue log ? We look at the disks centered at the hexagon



Hence, the domain is not "fitting" inside $\mathbb{C}$, so there is a *need* to go beyond the complex plane...

We may describe the *algorithm*

---

cocycle condition

$$f(z) = \sqrt{z} = (1 + z - 1)^{1/2}$$

write binomial series

$$(1+u)^a = 1 + au + \frac{a(a-1)}{2}u^2 + \dots$$

converges for $|u| < 1$

$w^2 = z^3 - z$

Luroth's theorem

Global holomorphic functions

> **Quote**
>
> **Riemann:** What is the *domain of definition* of an holomorphic function?

Let the union of all germs of holomorphic functions ("*Etale space*") be

$$\mathscr{O} := \left\{(a, f) : a \in \mathbb{C},\ f(z) = \sum_{n \geq 0} c_n (z-a)^n\right\}$$

and we want a topology on this such that

$$\begin{aligned} \mathscr{O} &\to \mathbb{C} \\ (a, f) &\mapsto a \end{aligned}$$

is a local homeomorphism.

We define

$$\begin{aligned} s_f : a + r_a D &\to \mathscr{O} \\ b &\mapsto (b, T_b f) \end{aligned}$$

where given a $(a, f) \in \mathscr{O}$ we have

$$(T_b f)(z) := \sum_{n \geq 0} f^n(b) \frac{z-b}{n!}$$

Then the images are

$$s_f(a + r_a D) = \{(b, g) \mid\}$$

> **Proposition:**
>
> $$\{s_f(a + r_a D) : (a, f) \in \mathscr{O}\}$$
>
> becomes a sub-basis for a topology on $\mathscr{O}$.

Now

$$\mathscr{O}$$

is the space of all global holomorphic functions with (domain points from $\mathbb{C}$ and valued in $\mathbb{C}$).

On this topological space we have two maps

- the projection π onto $\mathbb{C}$
- the *evaluation* of the function

$$\begin{tikzcd} & \begin{matrix}\mathscr{O}\\(a, f)\end{matrix} \arrow[r, "\mathrm{ev}"] \arrow[d, "\pi"'] & \begin{matrix}\mathbb{C}\\f(a)\end{matrix} \\ a + r_a D \arrow[ur, "s_f"] \arrow[r, hook] & \begin{matrix}\mathbb{C}\\a\end{matrix} & \end{tikzcd}$$

Definition 1. We define the *derivative* d, a map $d\colon \mathcal{O} \to \mathcal{O}$ as follows: If $f_a \in \mathcal{O}_a$ and (U, f) is a representative of f_a, we define $d(f_a) = df_a$ to be the germ at a of (U, f') where $f' = df/dz$ is the derivative of f.

Proposition 1. *$d\colon \mathcal{O} \to \mathcal{O}$ is a covering map.*

Proof. Let $f_a \in \mathcal{O}$ and let (U, f) be a representative of f_a. Let D be a disc centered at a such that $D \subset U$.

Let F be a primitive of f on D, and let $\mathcal{D} = N(D, f)$. For any $c \in \mathbb{C}$, let $U_c = N(D, F + c)$. We claim that $d^{-1}(\mathcal{D}) = \bigcup_{c\in\mathbb{C}} U_c$.

To prove this, let $z \in D$, $g_z \in \mathcal{O}_z$ and $dg_z = f_z$. Let (W, g) be a representative of g_z where W is a connected neighborhood of z, $W \subset D$. Then $g' = f$ in a neighborhood of z, hence $g' = f$ on W so that $(d/dz)(g - F) = 0$ on W. Hence, for some $c \in \mathbb{C}$, $g = F + c$ on W and $g_z \in U_c$. Conversely, we trivially have $dU_c = N(D, f) = \mathcal{D}$.

We now check that $d|U_c$ is a homeomorphism onto $\mathcal{D}$ for any $c \in \mathbb{C}$. We have only to check that $d|U_c$ is injective, which is obvious since d takes distinct elements of U_c to germs at different points of D. Since, finally, the U_c with distinct values of c are mutually disjoint, $\mathcal{D}$ is evenly covered by d.

We shall now show that there is a close connection between integration along curves and the lifting of curves relative to the derivative $d\colon \mathcal{O} \to \mathcal{O}$.

connected components

A connected component of $\mathscr{O}$ is a "Riemann surface" S. There are two functions on it

$$S \to \mathbb{C}; (a, f) \mapsto a$$

$$S \to \mathbb{C}; (a, f) \mapsto f(a)$$

- a $(0, f)$ whose connected component only contains disk?

constructing global holomorphic functions

- The complex logarithm
- Radicals $z^{1/k}$
- Inverse of a holomorphic function $f : U \to \mathbb{C}$
- Primitive if a 1-form $f(z)\mathrm{d}z$ on $U \subseteq \mathbb{C}$
- Radical $\sqrt[k]{f}$ of a holomorphic function $f : U \to \mathbb{C}$

adding the missing points: meromorphic functions on $\mathbb{C}P^1$

The corresponding space for meromorphic functions on $\mathbb{C}P^1$ is

$$\begin{aligned}\mathscr{M} := &\left\{(a, f) \middle| f = \sum_{n=-N}^{\infty} c_n (z-a)^n, \limsup_n |c_n|^{1/n} < \infty\right\} \\ &\cup \left\{(\infty, f) \middle| f(z) = \sum_{n=-N}^{\infty} c_n \frac{1}{z^n}, \limsup_n |c_n|^{1/n} < \infty\right\}\end{aligned}$$

So

$$\mathscr{O} \subset \mathscr{M}$$

but its larger because

$$\left(0, \frac{1}{z}\right) \in \mathscr{M}$$

We extend the topology on $\mathscr{O}$ to a topology on $\mathscr{M}$.

We have the extended maps

$$\begin{array}{ccc} \mathscr{M} & \xrightarrow{\text{ev}} & \mathbb{C}P^1 \\ (a, f) & & f(a) \\ \pi \downarrow & & \\ \mathbb{C}P^1 & & \\ a & & \end{array}$$

where

> **Proposition:** π is a local homeomorphism.

💡 (check for the points at infinity...)

adding the missing points: finite ramifications

Consider rD^* and its inverse image in $\mathscr{M}$. There are many connected components of the pre-image. We consider a component $E \subset \mathscr{M}$ such that

$$\pi : E \to rD^*$$

is **proper**.

These exist.

And now by

> **Proposition:**
>
> 11-9. Show that a proper local homeomorphism between connected, locally path-connected, and compactly generated Hausdorff spaces is a covering map.
>
> 11-10. Show that a covering map is proper if and only if it is finite-sheeted.

we know

$$\pi : E^* \to rD^*$$

is a finite covering map of the puntchered disk!

They are classified by finite subgroups $\pi_1(D^*) \cong \mathbb{Z}$, and they are of the form:

$$D^* \to D^*, z \mapsto z^n$$

or

$$\mathscr{O}[z^{1/n}] \times_{\mathbb{C}^*} D^* = D^*_{1/n} =: \pi^{-1}(D^*) \xrightarrow{\pi} D^*$$

which gives

$$\begin{array}{ccccc} & & E^* & \longrightarrow & D^* \\ & \swarrow^{f} & \uparrow & \nearrow & \\ \mathbb{C}P^1 & \longleftarrow & D^*_{1/n} & \hookrightarrow & D_{1/n} \end{array}$$

$$\prod_i (w - f \circ \gamma^i) = w^n + a_1 w^{n-1} + \cdots + a_n$$

then these a_i are holomorphic functions well defined on D^* which has a removable singularity at 0, so they extend onto D. Now by

(Newton-Puiseux series)

[1]

we can solve for f on $D_{1/n}$.

Now we add these points

$$(0, -)$$

...

and we construct

$$\overline{\mathscr{M}}$$

adds onto $\mathscr{M}$ the "punctures" to the punctured disks in $\mathscr{M}$ that occurs as finite covers of punctured disks in $\mathbb{C}P^1$.

The components of $\overline{\mathscr{M}}$ solves the problem of finding the maximal domain of definition of a meromorphic function on $\mathbb{C}P^1$.

---

1. Walker book ↩

$\sum_{n\geq 1} z^{n!}$

[1]

$$z + z^{2!} + z^{3!} + \dots$$

- On $K(\text{compact}) \subset D$, $|z| = r < 1$, then

$$\left|\sum_{n\geq 1} z^{n!}\right| \leq \sum_{n\geq 0} r^n = \frac{1}{1-r} < \infty$$

- Given a rational $\frac{p}{q}, q \geq 1$ [2] [3] for $z = r\exp\left(i\frac{\pi p}{q}\right)$ we have

$$\sum_{n\geq 1}\left(r\exp\left(i\pi\frac{p}{q}\right)\right)^{n!} = \sum_{n\geq 1} r^{n!}\exp\left(i\pi n!\frac{p}{q}\right)$$

    - ...

Proposition: The function

$$\begin{aligned} D &\to \mathbb{C} \\ z &\mapsto \sum_{n\geq 0} z^{n!} \end{aligned}$$

has the unit circle S^1 as its circle of convergence. It is unbounded in every neighborhood of every point on its circle of convergence, thus the unit circle is the natural boundary of the function.

$g(z) = \sum_{n\geq 0} \frac{z^{n!}}{n!}$

$$zg'(z) = \sum_{n\geq 1} z^{n!}$$

---

1. math.ualberta.ca/~isaac/math311/s14/lac.pdf ↩
2. https://math.stackexchange.com/a/2156862 ↩
3. https://math.stackexchange.com/a/4985000 ↩

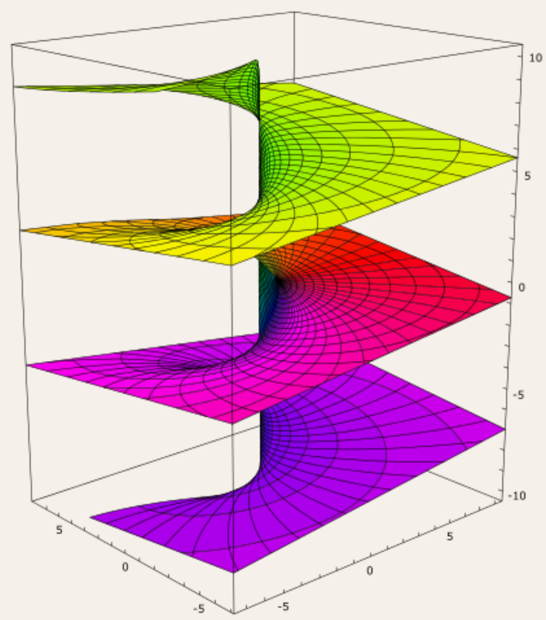
$\mathbb{Z}$-sheeted cover of punctured plane $U(\mathbb{R}^2 \setminus \{(0,0)\})$

#wiki/Man/R/2

We construct the connected, simply connected, 2 dimensional smooth $\mathbb{R}$-manifold (diffeomorphic to $\mathbb{R}^2$) that is the universal cover of $\mathbb{R}^2 \setminus \{(0,0)\}$.

Definition.

$$\mathbb{Z} \curvearrowright (U(\mathbb{C}^\times), +) \xrightarrow{\pi} (\mathbb{C}^\times, \times)$$



Consider

$$\mathbb{C} \setminus \{0\}$$

and

$$U(\mathbb{C} \setminus \{0\}) := \frac{\mathbb{C} \setminus (-\infty, 0] \times \mathbb{Z} \coprod (-\infty, 0) \times (-1, 1) \times \mathbb{Z}}{(z, k)}$$

- by taking

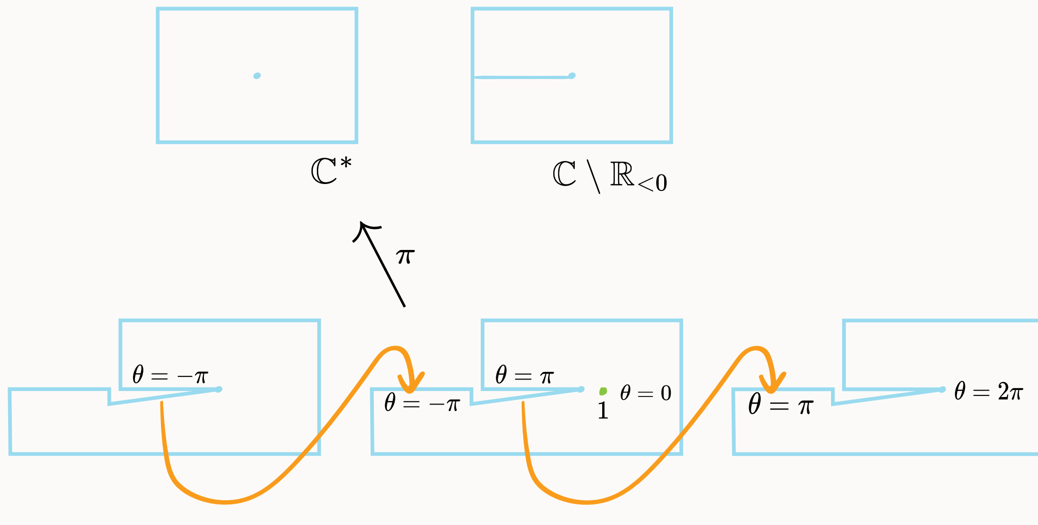
$$\coprod_{\mathbb{Z}} \mathbb{R}^2 \setminus \{(0, x) | x \leq 0\}$$

  and gluing the edge of the n-th piece with the edge of $n+1$-th piece.
  - The elements look like

$$(x, y, \theta)$$

    where θ is defined by adding $2\pi n$ to $\tan^{-1}(y/x)$.
  - This forms simply connected (smooth) cover of $\mathbb{R}^2 \setminus \{0\}$ with covering map

$$\begin{aligned} \pi : U(\mathbb{C}^\times) &\to \mathbb{C}^\times \\ (x, y, \theta) &\mapsto (x, y) \end{aligned}$$



$U(\mathbb{C}^*)$

$\mathbb{C}$, where elements look like (z, θ) whose complex structure is given by the covering map π.

- We lift the group structure

$$\begin{aligned}&(U(\mathbb{C}^\times), +)\\ &(z_1, \theta_1) + (z_2, \theta_2) = (z_1 z_2, \theta_1 + \theta_2)\end{aligned}$$

  from $(\mathbb{C}^\times, \times)$ such that π is a group homomorphism by choosing $(1, 0)$ as the identity.

  - Already a choice has been made for the identity on the cover, now we let the two identity to be base points, we have a canonical lift

$$\begin{aligned}\mathbb{C}^\times \setminus \mathbb{R}_{<0} &\hookrightarrow U(\mathbb{C}^\times)\\ z = x + iy &\mapsto \tilde{z} := (z, \tan^{-1}(y/x))\end{aligned}$$

    such that $\pi^{-1}(z) = \tilde{z} + (1, 2\pi i\mathbb{Z})$

  - We can now write

$$(z, \theta) = \tilde{z} + (1, 2\pi n)$$

- With the covering group action by $\mathrm{Deck}(U(\mathbb{C}^\times) \to \mathbb{C}^\times) \cong \pi_1(\mathbb{C}^\times) \cong \mathbb{Z}$ is generated by σ given by

$$z \overset{\sigma}{\mapsto} z + (1, 2\pi i)$$

$U(\mathbb{C}^\times)$ is a Riemann surface biholomorphic to $\mathbb{C}$ which is the complex Lie group covering of $\mathbb{C}^\times$. The map

$$\theta : U(\mathbb{C}^\times) \to \mathbb{R}$$

is smooth and surjective.

embedding inside $\mathbb{R}^3$

$$\begin{aligned}U(\mathbb{R}^2 \setminus \{0\}) &\hookrightarrow \mathbb{R}^3\\ (x, y, \theta) &\mapsto (x, y, \theta)\end{aligned}$$

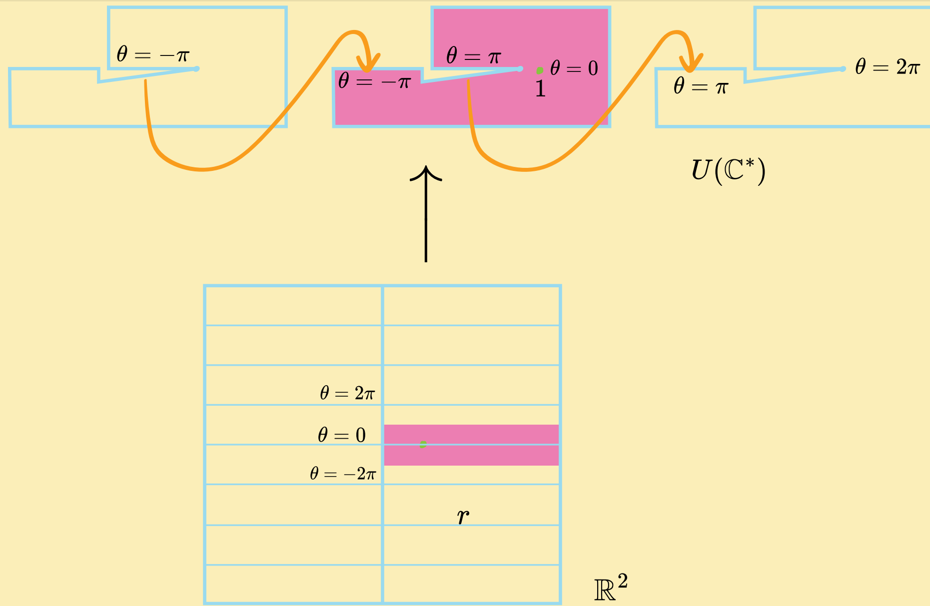
The image is a subset of

$$\mathbb{R}^2 \setminus \{0\} \times \mathbb{R}$$

polar coordinates actually cover the universal cover!

The "polar coordinates on $\mathbb{R}^2 \setminus \{0\}$" actually covers the infinite helicoid

$$\begin{aligned}\mathbb{R}_{>0} \times \mathbb{R} &\to U(\mathbb{C}^\times)\\ (r, \theta) &\mapsto (r\cos\theta, r\sin\theta, \theta)\end{aligned}$$



is a diffeomorphism.

which in complex coordinates this is

$$\begin{aligned} \mathbb{C}_{\mathrm{Re}>0} &\to U(\mathbb{C}^\times) \\ (r, \theta) &\mapsto (r\exp(i\theta), \theta) \end{aligned}$$

but it is of course **not** a bi-holomorphism?

A better way to think about this is: we think of

$$\begin{aligned} \mathbb{C}_{\mathrm{Re}>0} &\to \mathbb{C}^\times \\ (r, \theta) &\mapsto r\exp(i\theta) \end{aligned}$$

as the covering map π itself, where we think of the right half plane *as* the universal cover, (but still this is not holomorphic?).

$$\begin{tikzcd} \begin{matrix}(0,\infty)\times\mathbb{R} \\ (|z|,\theta)\end{matrix} \arrow[rr, bend left] \arrow[dr] & & \begin{matrix}U(\mathbb{C}\setminus\{0\}) \\ (z,\theta)\end{matrix} \arrow[ll, bend left] \arrow[dl] \\ & \begin{matrix}(0,\infty)\times S^1 \cong \mathbb{C}\setminus\{0\} \\ |z|e^{i\theta} = z\end{matrix} & \end{tikzcd}$$

We need

$$\exp : \mathbb{C} \to \mathbb{C}^\times$$

as the holomorphic map, which is the covering map.

the complex logarithm

complex logarithm on the universal cover of punctured plane

The $U(\mathbb{C}^\times)$ is the Riemann surface of $\log(z)$

$$\begin{aligned} \log : U(\mathbb{C}^\times) &\to \mathbb{C} \\ (z, \theta) &\mapsto \log_{\mathbb{R}}(|z|) + i\theta \end{aligned}$$

$$
\begin{tikzcd}
\begin{matrix} U(\mathbb{C}\setminus\{0\}) \\ (z,\theta) \end{matrix} \arrow[rr, bend left] \arrow[dr] & & \begin{matrix} \mathbb{C} \\ ? \end{matrix} \arrow[ll, bend left] \arrow[dl, "\exp"] \\
& \begin{matrix} \mathbb{C}\setminus\{0\} \\ z = \exp(w) \end{matrix} &
\end{tikzcd}
$$

$$
z = \exp(f(z,\theta))
$$

the pullback of inexact closed forms on punchered plane

$$
H^1(\Omega^\bullet(\mathbb{R}^2\setminus\{0\})) = \mathbb{R}\left[\frac{-y\mathrm{d}x + x\mathrm{d}y}{x^2+y^2}\right]
$$

$$
\pi^*\frac{-y\mathrm{d}x + x\mathrm{d}y}{x^2+y^2} = \mathrm{d}\theta
$$

The usual definition of Riemann surfaces

A smooth $\mathbb{R}$-manifold has a complicated definition:

Definition. **A smooth $\mathbb{R}$-manifold is a** $\mathcal{C}^\infty$-manifold $(M, \mathcal{T}_M, \mathcal{A}_\infty^{\max})$

A **$\mathcal{C}^\infty$-manifold** is a triple $(M, \mathcal{T}_M, \mathcal{A}_\infty^{\max})$ where

- $(M, \mathcal{T}_M)$ is a

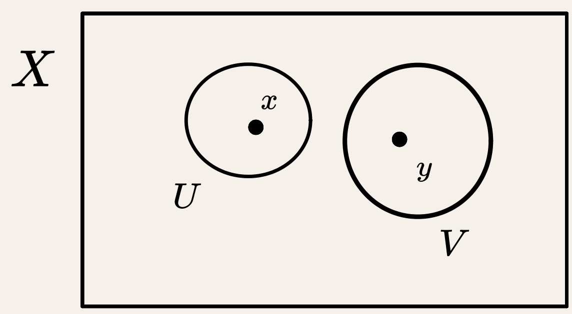
  Definition. **Topological manifold**

  A topological space is a real topological n-manifold or **$(\mathbb{R}^n, \mathcal{C}^0)$-manifold** for a $n \in \mathbb{Z}^+$ (or topological n-manifold) if

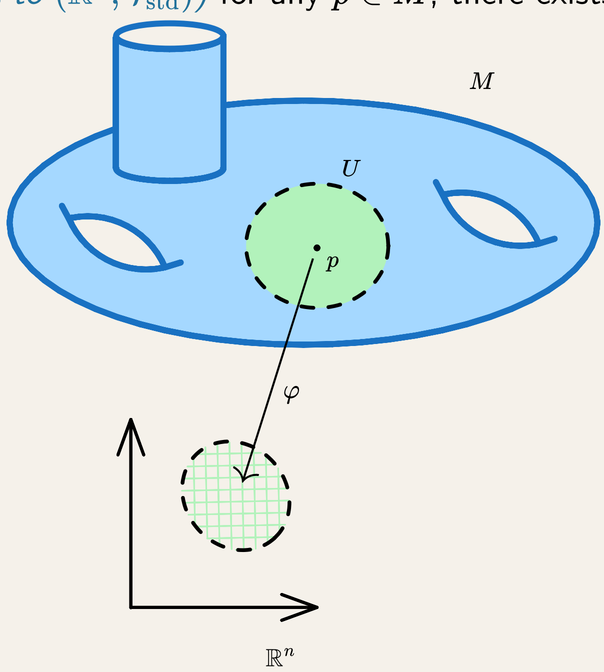
  - M is Hausdorff topological space

    Definition. **T_2 topological space or Hausdorff space**

    A topological space is said to be **T_2 or Hausdorff** if for any pair $x, y \in X$, $\exists U, V \in \mathcal{T}$ such that $x \in U$, $y \in V$, and $U \cap V = \emptyset$.

  - M is second countable
  - *(locally homeomorphic to $(\mathbb{R}^n, \mathcal{T}_{\text{std}})$)* for any $p \in M$, there exists

    - an open subset $U \subseteq M$ with $p \in U$
    - open subset $V \subseteq \mathbb{R}^n$
    - and a homeomorphism $\varphi : U \to V$

- $\mathcal{A}_\infty^{\max}$ is a maximal $\mathcal{C}^\infty$-atlas

  Definition. **$\mathcal{C}^\bullet$-compatible charts**

  Two charts (U, x) and (V, y) are said to be $\mathcal{C}^\bullet$-compatible if either

- $U \cap V = \emptyset$ or
- the map $y \circ x^{-1}$ is $\mathcal{C}^\bullet$
    - $\bullet = 0$
    - $\bullet = k \in \mathbb{Z}^+$
    - $\bullet = \infty$
    - $\bullet = \omega$

A **$\mathcal{C}^\bullet$-atlas** of a topological manifold is an atlas of pairwise $\mathcal{C}^\bullet$-compatible charts. We denote it by $\boldsymbol{\mathcal{A}_\bullet}$

A $\mathcal{C}^\bullet$-atlas $\boldsymbol{\mathcal{A}_\bullet}$ of a topological manifold is a **maximal $\mathcal{C}^\bullet$-atlas** if for every $(U, x) \in \boldsymbol{\mathcal{A}_\bullet}$ we have all the charts $(V, y) \in \boldsymbol{\mathcal{A}_\bullet}$ that are $\bullet$-compatible with (U, x).
We denote it by $\boldsymbol{\mathcal{A}}_\bullet^{\max}$

We need the maximal atlas to properly give the topological space a "smooth structure" that "makes sense".

Now *complex manifolds* are easy to describe:

Definition. $\mathbb{C}$-manifolds

Let M be a smooth $\mathbb{R}$-manifold of even $\mathbb{R}$-dimension $2n$.
A smooth atlas on a smooth $\mathbb{R}$-manifold is **homomorphic** $\mathcal{A}_{\mathscr{O}}$ if for any pair of charts, the transition functions are *biholomorphic*

$$(U, z), (V, w) \in \mathcal{A}_{\mathscr{O}} \implies z \circ w^{-1} \in \mathscr{O}$$

It determines a maximal holomorphic atlas $\mathcal{A}_{\mathscr{O}}^{\max}$ *holomorphic (or complex) structure*. Then

$$(M, \mathcal{T}_M, \mathcal{A}_{\mathscr{O}})$$

is a complex $\mathbb{C}$-manifold of $\mathbb{C}$-dimension 1.

With all this *language* setup, a Riemann surface becomes a simple object:

Definition. Riemann surfaces

A **Riemann surface** is a connected $\mathbb{C}$-manifold of $\mathbb{C}$-dimension 1.

Holomorphic maps between Riemann surfaces are defined in the same way as manifolds:

Definition. Holomorphic maps between Riemann surfaces

Let

$$f : X \to Y$$

be a man between Riemann surfaces X, Y. Then is **holomorphic** if for pair of charts $(U, \varphi) \ni P$ and $(V, \psi) \ni f(P)$ for X, Y respectively the induced map

$$\psi \circ f \circ \varphi^{-1} : \varphi(U) \to \psi(V)$$

is *holomorphic*.

Holomorphic functions on a complex 1-manifold

If X is a compact connected Riemann surface, then

$$\mathcal{O}(X) \cong \mathbb{C}$$

Take any holomorphic $f \in \mathcal{O}(X)$ then

- $|f|$ is a continuous function from compact to $\mathbb{R}$, so there exists a maximum, that is, there exists $x_0 \in X$ such that

$$|f(x_0)| = \sup\{|f(x)| : x \in X\}$$

- $f^{-1}(x_0)$ is a closed subset of X
- take $x \in f^{-1}(x_0)$ and a chart nbd $U\alpha$ of x and move $f\mid_{U\alpha}$ onto a function between subsets of $\mathbb{C}$, then because modulus of non-constant holomorphic functions cannot attain its maximum in the interior of its domain, it must be a constant on U_α, which is a open nbd of x. So every element of $f^{-1}(x_0)$ has a open nbd inside it. So $f^{-1}(x_0)$ is **open**

Because X is connected, $f^{-1}(x_0) = X$. Thus f is constant on X.

Differential forms on complex 1-manifolds

These are smooth complex differential forms on a Riemann surface X

$$\begin{array}{ccccc}
\Omega^0(X,\mathbb{C}) & \to & \Omega^{1,0}(X,\mathbb{C}) \oplus \Omega^{0,1}(X,\mathbb{C}) & \to & \Omega^2(X,\mathbb{C}) \\
\phi & \overset{\mathrm{d}}{\mapsto} & \mathrm{d}\phi = \frac{\partial\phi}{\partial z}\mathrm{d}z + \frac{\partial\phi}{\partial\bar{z}}\mathrm{d}\bar{z} & & \\
 & & f\mathrm{d}z + g\mathrm{d}\bar{z} & \overset{\mathrm{d}}{\mapsto} & \left(\frac{\partial g}{\partial z} - \frac{\partial f}{\partial\bar{z}}\right)\mathrm{d}z \wedge \mathrm{d}\bar{z} \\
 & & & & h\mathrm{d}z \wedge \mathrm{d}\bar{z}
\end{array}$$

These are $\mathcal{C}^\infty(M,\mathbb{C})$-differential forms on a Riemann surface and the exterior derivative.

The Hodge diamond of Riemann surfaces

$$\begin{array}{ccc}
 & h^{0,0} & \\
h^{1,0} & & h^{0,1} \\
 & h^{1,1} &
\end{array}$$

Hodge diamonds

Proposition:

- $$H^{0,0} = \frac{\Omega^{0,0} \cap \ker\bar{\partial}}{0} = \Omega^0_{\mathscr{O}} = \mathscr{O}$$
- $$H^{0,1} = \frac{\Omega^{0,1}}{\bar{\partial}\Omega^{0,0}} = 0$$
- $$H^{1,0} = \frac{\Omega^{1,0} \cap \ker\bar{\partial}}{0} = \Omega^1_{\mathscr{O}}$$
- $$H^{1,1} = \frac{\Omega^{1,1}}{\bar{\partial}\Omega^{1,0}} = 0$$

Proposition:

- For a connected **compact** Riemann surface, the Hodge diamond looks like

$$\begin{array}{ccccccc}
 & h^{0,0} & & & & 1? & \\
h^{1,0} & & h^{0,1} = g & & g & & g \\
 & h^{1,1} & & & & 0? &
\end{array}$$

- For a connected **non-compact** Riemann surface, the Hodge diamond looks like

$$\begin{array}{ccccccc}
 & h^{0,0} & & & & \infty? & \\
h^{1,0} & & h^{0,1} = \infty? & & & & 0 \\
 & h^{1,1} & & & & 0 &
\end{array}$$

Holomorphic forms

Proposition:

- For a connected **non-compact Riemann surface** M

$$H^1(\Omega^\bullet_{\mathscr{O}}(M)) = \frac{\Omega^1_{\mathscr{O}}}{\mathrm{d}\Omega^0_{\mathscr{O}}} \cong H^1(M, \mathbb{C})$$

-

Theorem 1.4.2. *If F is meromorphic in a neighborhood of ζ, then there is a neighborhood of ζ, where*

$$F(z) = \sum_{1}^{n} A_k(z - \zeta)^{-k} + G(z)$$

with constants A_k and an analytic function G. The representation is unique. If $F_\zeta \neq 0$ there is also a unique representation of the form

$$F(z) = (z - \zeta)^n G(z),$$

where $G(\zeta) \neq 0$ and n is an integer. If $n > 0$, we have a zero of order n at ζ; and if $n < 0$, we have a pole of order $-n$.

The proof is an obvious consequence of Corollary 1.2.10. We shall next discuss the first representation; the multiplicative representation will be studied in section 1.5.

Theorem 1.4.3. (*Mittag-Leffler*) *Let z_j, $j = 1, 2, \cdots$, be a discrete sequence of different points in the open set Ω, and let f_j be meromorphic in a neighborhood of z_j. Then there exists a meromorphic function f in Ω such that f is analytic outside the points z_j and $f - f_j$ is analytic in a neighborhood of z_j for every j.*

Proof. In view of Theorem 1.4.2 we may assume that

$$f_j(z) = \sum_{1}^{n_j} A_{jk}(z - z_j)^{-k}.$$

We want to find functions $u_j \in A(\Omega)$ so that the series

$$f(z) = \sum_{1}^{\infty} (f_j(z) - u_j(z))$$

defines a function f with the required properties. To do so we choose an increasing sequence of compact sets $K_j \subset \Omega$ with $\hat{K}_j = K_j$ so that every compact subset of Ω is contained in some K_j. We may assume that $z_k \notin K_j$ when $k \geq j$, since the points z_k have no accumulation point in Ω. By the Runge theorem we can then choose $u_j \in A(\Omega)$ so that

$$|f_j(z) - u_j(z)| < 2^{-j}$$

in K_j. But then the series

$$\sum_k^{\infty} (f_j(z) - u_j(z))$$

converges uniformly on K_k to a function which is analytic in the interior of K_k. Hence the definition of f above is meaningful and f has the required properties.

Another formulation of the Mittag-Leffler theorem is the following, which must be used in the case of several variables:

Theorem 1.4.3′. *Let $\Omega = \cup_j \Omega_j$ where Ω_j are open sets in $\mathbf{C}$. If $f_j \in M(\Omega_j)$ and $f_j - f_k \in A(\Omega_j \cap \Omega_k)$ for all j and k, one can find $f \in M(\Omega)$ so that $f - f_j \in A(\Omega_j)$ for every j.*

That this is equivalent to Theorem 1.4.3 may be verified by the reader. Another equivalent result is the following:

Theorem 1.4.4. *For every $f \in C^\infty(\Omega)$ the equation $\partial u/\partial \bar{z} = f$ has a solution $u \in C^\infty(\Omega)$.*

Proof. Choose an increasing sequence of compact sets $K_j \subset \Omega$ with $\hat{K}_j = K_j$ so that every compact subset of Ω is contained in some K_j. Let $\psi_j \in C_0^\infty(\Omega)$ be equal to 1 in a neighborhood of K_j and set $\varphi_1 = \psi_1$, $\varphi_j = \psi_j - \psi_{j-1}$ when $j > 1$. Then $\varphi_j = 0$ in a neighborhood of K_{j-1} and $\Sigma_1^\infty \varphi_j = 1$ in Ω. By Theorem 1.2.2 we can find $u_j \in C^\infty(\mathbf{R}^2)$ so that $\partial u_j/\partial \bar{z} = \varphi_j f$. This means in particular that u_j is analytic in a neighborhood of K_{j-1}. By the Runge theorem we can therefore choose $v_j \in A(\Omega)$ so that $|u_j - v_j| < 2^{-j}$ in K_{j-1}. Then the sum

$$u = \sum_1^\infty (u_j - v_j)$$

is uniformly convergent on every compact set in Ω. The sum from $l + 1$ to ∞ consists of terms which are analytic near K_l and it converges uniformly on K_l to a function which is analytic in the interior of K_l. Hence $u \in C^\infty(\Omega)$, and since $\partial u/\partial \bar{z}$ can be computed by termwise differentiation, we have

$$\partial u/\partial \bar{z} = \sum_1^\infty \varphi_j f = f.$$

This completes the proof.

We shall now show that Theorem 1.4.4 implies a strengthened form of Theorem 1.4.3′.

Theorem 1.4.5. *Let $\Omega = \cup_1^\infty \Omega_j$ and let $g_{jk} \in A(\Omega_j \cap \Omega_k)$, $j, k = 1, 2, \cdots$ satisfy the conditions*

$$g_{jk} = -g_{kj}, \quad g_{jk} + g_{kl} + g_{lj} = 0 \quad \text{in } \Omega_j \cap \Omega_k \cap \Omega_l \text{ for all } j, k, l. \tag{1.4.1}$$

Then one can find $g_j \in A(\Omega_j)$ so that

$$g_{jk} = g_k - g_j \quad \text{in } \Omega_j \cap \Omega_k \text{ for all } j \text{ and } k. \tag{1.4.2}$$

Proof that Theorem 1.4.5 implies Theorem 1.4.3′. With the notations of Theorem 1.4.3′ we set $g_{jk} = f_j - f_k$. The hypothesis (1.4.1) of Theorem 1.4.5 is then fulfilled, so we can find $g_j \in A(\Omega_j)$ satisfying the equations

$$f_j - f_k = g_{jk} = g_k - g_j \quad \text{in } \Omega_j \cap \Omega_k \text{ for all } j \text{ and } k.$$

This means that $f_j + g_j = f_k + g_k$ in $\Omega_j \cap \Omega_k$. Hence there is a meromorphic function f in Ω such that $f = f_j + g_j$ in Ω_j for every j and, since $f - f_j = g_j \in A(\Omega_j)$, this proves Theorem 1.4.3′.

Proof of Theorem 1.4.5. We can choose a partition of unity subordinate to the covering $\{\Omega_j\}$, that is, we can choose functions φ_ν and positive integers i_ν, $\nu = 1, 2, \cdots$ so that

(i) $$\varphi_\nu \in C_0^\infty(\Omega_{i_\nu}).$$

(ii) All but a finite number of functions φ_ν vanish identically on any compact subset of Ω.

(iii) $$\sum \varphi_\nu = 1 \quad \text{on } \Omega.$$

(See for example Schwartz [1] or de Rham [1].)

If (1.4.2) is fulfilled, we obtain by taking $j = i_\nu$, multiplying by φ_ν and adding

$$g_k = h_k + u,$$

where we have written

$$h_k = \sum \varphi_\nu g_{i_\nu k}$$

and $u = \Sigma\, \varphi_\nu g_{i_\nu}$. The function h_k is well defined in terms of the functions g_{jk} (we set $\varphi_\nu g_{i_\nu k} = 0$ outside Ω_{i_ν}) and belongs to $C^\infty(\Omega_k)$. Furthermore, we have

$$h_k - h_j = \sum \varphi_\nu (g_{i_\nu k} - g_{i_\nu j}) = \sum \varphi_\nu g_{jk} = g_{jk}$$

since $g_{ki_\nu} + g_{i_\nu j} + g_{jk} = 0$. This implies that

$$\partial h_k / \partial \bar z = \partial h_j / \partial \bar z \quad \text{in } \Omega_j \cap \Omega_k.$$

Hence there is a function $\psi \in C^\infty(\Omega)$ such that

$$\psi = \partial h_k/\partial \bar{z}, \quad \text{in } \Omega_k \text{ for every } k.$$

If we now choose u as a solution of the equation

$$\partial u/\partial \bar{z} = -\psi,$$

which is possible by Theorem 1.4.4, the functions $g_k = h_k + u$ have all the required properties.

Thus we see that statements like Theorem 1.4.4 or Theorem 1.4.5 or Theorem 1.4.3′ are essentially equivalent. In the case of several variables we shall in this book prefer to work with results similar to Theorem 1.4.4, postponing the proof of results like Theorem 1.4.5 to the last chapter where we shall give an appropriate terminology for their study (cohomology groups with values in a sheaf).

Concept of a Riemann surface: usual definition vs as a ringed space

Given X a topological space we want to "define holomorphic functions locally"

$$U \mapsto \mathscr{O}(U)$$

where $\mathscr{O}(U)$ is the commutative $\mathbb{C}$-algebra of holomorphic functions on $U \subset X$. Consider

$$\{U_i\}_{i\in I}, V = \bigcup_{i\in I} U_i$$

and

$$f_i \in \mathscr{O}(U_i)$$

such that

$$f_i\Big|_{U_i\cap U_j} = f_j\Big|_{U_i\cap U_j}$$

for all $i, j \in I$, then there *exists an unique* $f \in \mathscr{O}(V)$ such that

$$f\Big|_{U_j} = f_j$$

Then $(X, \mathscr{O})$ is called a **ringed space**.

In the *abstract setting* this association is called a *sheaf*

Definition. Sheaf on open sets of a topological space

A *functor* from the opposite of the category of open sets on X

$$\mathscr{S} : (\mathrm{Open}(X), \subseteq)^{op} \to \mathsf{C}$$

where the morphism $U \subseteq V$ maps to the morphism

$$\mathrm{res}_{V,U} : \mathscr{S}(V) \to \mathscr{S}(U)$$

in C and when $\{U_i\}_{i\in I}$ is an open cover of $U \subseteq X$ then

- *(identity)*

$$f \in \mathscr{S}(U), \forall i \in I,\ \mathrm{res}_{U,U_i}(f) = 0 \implies f = 0$$

- *(gluability)* Given $f_i \in \mathscr{S}(U_i)$ for all $i \in I$, such that

$$\mathrm{res}_{U_i,U_i\cap U_j}(f_i) = \mathrm{res}_{U_j,U_i\cap U_j}(f_j)$$

  then

$$\begin{aligned} \exists f \in \mathscr{S}(U) \\ \mathrm{res}_{U,U_i}(f) = f_i \end{aligned}$$

Roughly speaking, sheaves are those functors, where local data can be glued to global data.

A ringed space is a topological space with a sheaf of rings on it. Usually ringed spaces are "modelled" on some well-known spaces, which means one of two things

- spaces modelled on $(\mathbb{C}^n, \mathscr{O})$ *locally*
- *globally*

Definition. Ringed space of functions modelled on $(D, \mathscr{O}|_D)$

A ringed space $(X, \mathscr{O})$ is called a **ringed space of functions modelled on $(D, \mathscr{O}|_D)$**

- if for each $p \in X$ there is a nbhd $p \in U$ such that

$$(U, \mathscr{O}|_U) \cong (D, \mathscr{O}_D)$$

- X is Hausdorff, second countable, connected
- $\mathscr{O}$ separates points

Add definition of morphisms of ringed spaces, and check if that is what we want above?

Question

Construct meromorphic functions on X above

$$\begin{tikzcd} X \arrow[r] \arrow[d] & \mathbb{C}P^1 \\ \mathbb{C}P^1 & \end{tikzcd}$$

such that there is a biholomorphism from X to a connected component in $\overline{\mathscr{M}}$ such that these two maps becomes evaluation and projection maps.

Riemann surfaces as analytic varieties of $\mathbb{C}P^n$ of dimension 1

We can also define Riemann surfaces as

Definition. Analytic varieties of $\mathbb{C}P^1$ of dimension 1

$$X \underbrace{\subset}_{\text{closed}} U \underbrace{\subset}_{\text{open}} \subset \mathbb{C}P^n$$

for some fixed $n \geq 3$ such that locally X is given by vanishing of holomorphic functions f_i such that

$$\left(\frac{\partial f_i}{\partial z_j}\right)$$

has rank $n-1$

and we must prove that these (three) definitions are equivalent.

Classification of complex 1-manifold

uniformization

(Uniformization of simply connected Riemann surfaces) Every *non-compact simply connected* Riemann surface is biholomprhic to the complex plane $\mathbb{C}$ or the unit disk D. Every *compact simply connected* Riemann surface is biholomorphic to the Riemann sphere $\mathbb{C}P^1$

overview

homeomorphism class	hol classes	moduli space	universal covering	Aut
plane $\mathbb{R}^2$	$\mathbb{C}$	$\{\bullet, \bullet\}$	$\mathbb{C}$	$\mathbb{C}$
	disk/upper half plane $D \cong H$		$D \cong H$	$PSL(2, \mathbb{R})$
cylinder $\cong \mathbb{R}^2 \setminus \{0\} \cong D \setminus \{(0,0)\}$	$\mathbb{C}$	$\{\bullet, \bullet\} \cup (0,1)$	$\mathbb{C}$	$\mathbb{C}$
	$D^* \cong H \setminus \{i\}$		$D \cong H$ by exponential map	$SO(2, \mathbb{R})$
	$D_r, r \in (0,1)$		$D \cong H$	??
sphere S^2	$\mathbb{C}P^1$	$\mathcal{M}_0 \cong \{\bullet\}$	$\mathbb{C}P^1$	$PGL(2, \mathbb{C})$
disk with $n > 1$ points removed		$\mathcal{M}_{0,n+1}$	$D \cong H$	?
torus T^2	a rank 2 lattice $\Lambda \subseteq \mathbb{C}$ upto homothety	$\mathcal{M}_1 \cong \dfrac{H}{SL(2, \mathbb{Z})} \cong \mathbb{C}$	$\mathbb{C}$ quotient by a rank 2 lattice	
higher genus surfaces $T_g^2, g > 1$		$\mathcal{M}_g \cong \mathbb{C}^{3g-3}$	$D \cong H$	finite
compact genus $g > 0$ surface with n points removed		$\mathcal{M}_{g,n}$	$D \cong H$ $D \cong H$	
compact genus g surface with infinite points removed			$D \cong H$	
infinite genus				
...				

Meromorphic functions on Riemann surfaces

Non-constant meromorphic functions on Riemann surfaces exist.

That is $\mathscr{M}(X)$ is not just $\mathbb{C}$.

There are two styles of proof of this.

Outline of proofs of

Non-constant meromorphic functions on Riemann surfaces exist.

local approaches

- "start with a holomorphc function on a small subset and somehow extend it to the whole Riemann surface"
- for

$$X = \bigcup_i U_i, V_i \subset U_i, V_i \cap \bar{U}_j = \emptyset$$

  use bumb functions

$$\phi_i \Big|_{V_i} = 1$$

  to construct a smooth function which is

$$g := \sum_i \phi_i \Big|_{V_i} = 0$$

  and then use *Perron's method* to increase the domain of holomorphicity

global approach: via uniformization

- for a Riemann surface X consider its universal cover $\tilde{X}$ which is a simply connected Riemann surface which is biholomorphic to one of three

  (Uniformization of simply connected Riemann surfaces) Every *non-compact simply connected* Riemann surface is biholomprhic to the complex plane $\mathbb{C}$ or the unit disk D. Every *compact simply connected* Riemann surface is biholomorphic to the Riemann sphere $\mathbb{C}P^1$

- now construct Deck transformation-invariant meromorphic functions on $\tilde{X}$
  - for $\mathbb{C}P^1$
  - for $\mathbb{C}$
    - $$\sum_{n \in \mathbb{Z}} \frac{1}{(z+n)^k}$$

      $k \geq 2$
  - for H

- *Poincare-Eistenstein series*

  $$\sum_{\gamma \in \Gamma} \frac{1}{(\gamma z)^k}$$

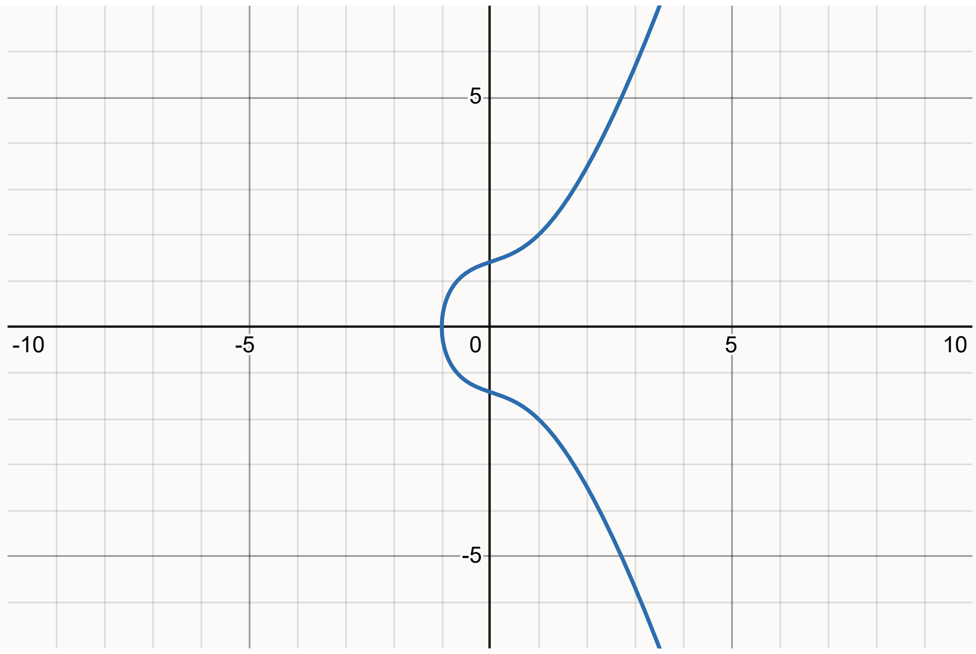
  for $k \geq 3$

- *factors of automorphy:* construct meromorphic functions of the form

$$f(\gamma z) = J(z, \gamma) f(z)$$

MTH416 presentation on *periods of elliptic curves*

Date: 23 April, 2025
Speaker: Rupadarshi Ray



1: Elliptic curves are compact Riemann surfaces

Let

$$\lambda \in \mathbb{C} \setminus \{0, 1\}$$

Then we have an elliptic curve in *Legendre form*

$$E := V(y^2 - x(x-1)(x-\lambda)) \subseteq \mathbb{C}P^2$$

which is a **compact Riemann surface** because

Complex projective curves in $\mathbb{C}P^2$ (zero sections of homogeneous polynomials) are compact.

- *compactness of $\mathbb{C}P^2$:* As $\mathbb{C}P^2$ is continuous image of a sphere under the projection map

  $$S^{2(2+1)-1} \subseteq \mathbb{C}^{2+1} \to \mathbb{C}P^2$$

  we conclude $\mathbb{C}P^2$ is **compact**.
- We look at the homogeneous polynomial

  $$P : \mathbb{C}^3 \setminus \{0\} \to \mathbb{C}$$

  and quotienting both spaces by the scaling action of $\mathbb{C}^\times$ by which the polynomial descends to a continuous

  $$P : \mathbb{C}P^2 \to \frac{\mathbb{C}}{\mathbb{C}^\times}$$

  by homogeneity of P.
- Under the quotient

  $$\mathbb{C} \to \frac{\mathbb{C}}{\mathbb{C}^\times} = \{[0], [1]\}$$

  the preimage of $[0]$ is $\{0\}$ which is closed, so in the quotient topology $[0]$ is closed.
- Thus $V(P) = P^{-1}(\{0\})$ is closed subset of a compact space $\mathbb{C}P^2 \implies$ it is compact.

We know $E \subseteq \mathbb{C}P^2$ is a Riemann surface whose charts are given by one of the four functions

- $$\begin{aligned}\{u_3 \neq 0\} \subseteq E &\to \mathbb{C}\\ [u_1, u_2, u_3] &\mapsto x := \frac{u_1}{u_3}\\ y &:= \frac{u_2}{u_3}\end{aligned}$$

- $$\begin{aligned}\{u_2 \neq 0\} \subseteq E &\to \mathbb{C}\\ [u_1, u_2, u_3] &\mapsto v := \frac{u_1}{u_2}\\ w &:= \frac{u_3}{u_2}\end{aligned}$$

which satisfy

$$\begin{aligned}x &= \frac{v}{w}\\ y &= \frac{1}{w}\end{aligned} \implies v = \frac{x}{y}$$

on the intersection of their domain of definition.

If

$$y^2 = x^3 + ax^2 + bx$$

then

$$w = v^3 + av^2w + bvw^2$$

or

$$\frac{1}{w^2} = f\left(\frac{v}{w}\right)$$

2: charts on the Elliptic curve

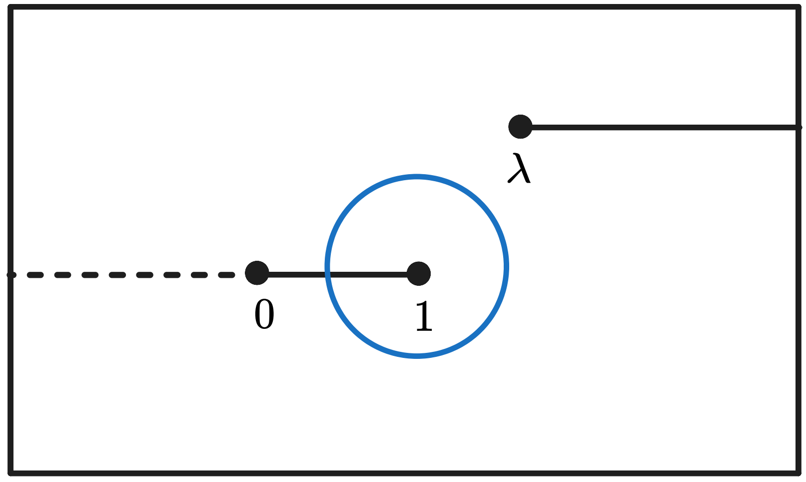
On any simply connected open subset of $\mathbb{C} \setminus \{0, 1, \lambda\}$ we may pick a branch of complex logarithm and define

$$\sqrt{f(x)} := \exp\left(\frac{1}{2}\log f(x)\right)$$

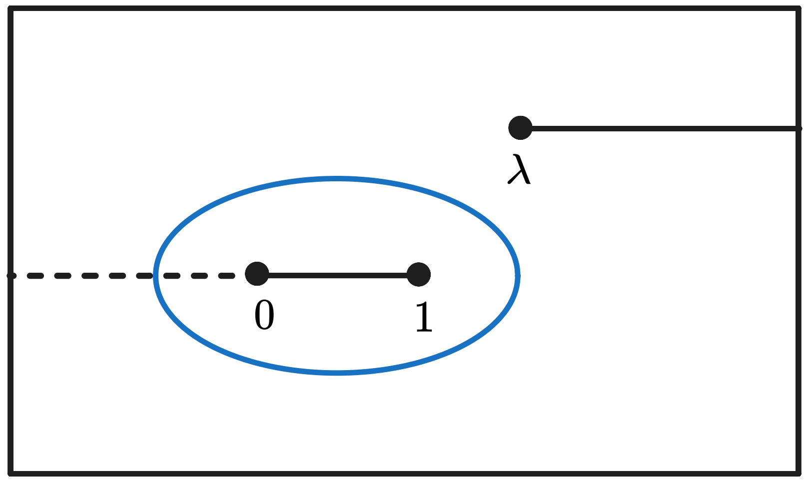
- The limit of the function $\sqrt{f}$ from above and below $[0, 1]$ do not match, as we may rephrase the limit in terms of limit of this branch of log around zero,

  $$\begin{aligned}\log z &\to \log z + 2\pi i\\ \implies \exp\frac{1}{2}\log z &\to \left(\exp\frac{1}{2}\log z\right)\exp(\pi i)\\ &= -\exp\left(\frac{1}{2}\log z\right)\end{aligned}$$

  so the function $\sqrt{f}$ "flips sign"



- However, going around $0, 1$ maps to going around the zero of $\log z$ twice, so the sign does not flip



  - So we can extend $\sqrt{f}$ to $(-\infty, 0)$.

Thus

> **Proposition:** For $S := [0,1] \cup L_\lambda$, there is a holomorphic square root of the function f which we denote by
>
> $$\sqrt{f} : \mathbb{C} \setminus S \to \mathbb{C}$$
>
> and $\sqrt{f} \neq 0$ on $\mathbb{C} \setminus S$.

So we have

$$\sqrt{f}, -\sqrt{f} : U_1, U_2 : \mathbb{C} \setminus S \to \mathbb{C}$$

are two branches of the function.

These functions define inverse of the charts on E_0

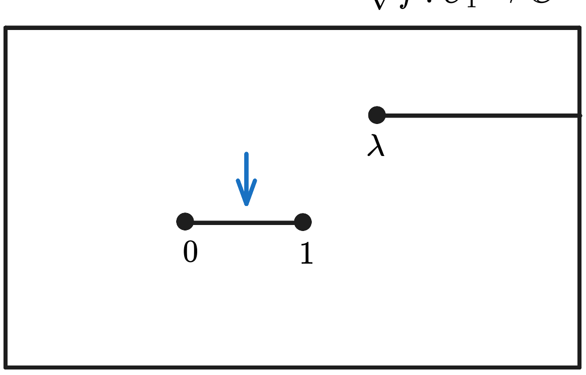
$$\begin{aligned} x : V_1 \subset E_0 &\leftrightarrow U_1 \\ (x, y) &\mapsto x \\ (x, \sqrt{f(x)}) &\leftarrow x \end{aligned}$$

and similarly for U_2.

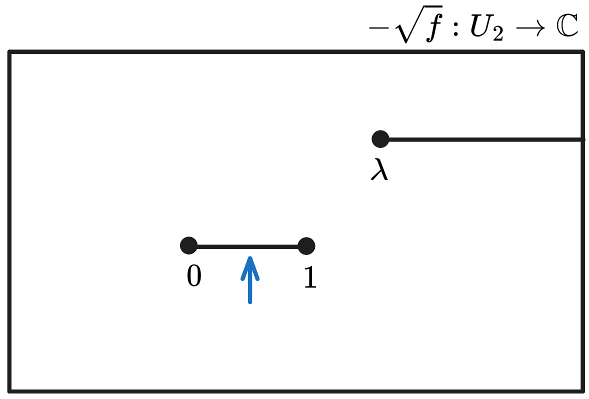
Riemann surface of $\sqrt{f}$

We observe the limit of $\sqrt{f}$ from above is equal to the limit of $-\sqrt{f}$ from below:

$\sqrt{f} : U_1 \to \mathbb{C}$



$-\sqrt{f} : U_2 \to \mathbb{C}$

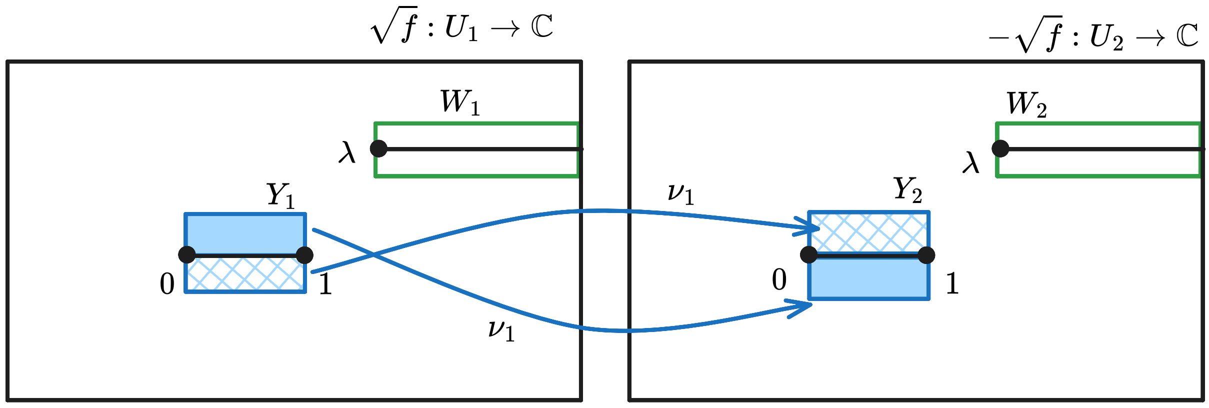


So we take two neighborhoods

$$\begin{aligned} Y_1 &= [0,1] \times (-1,1) \subset U_1 \\ Y_2 &= [0,1] \times (-1,1) \subset U_2 \end{aligned}$$

and define the map that reflects the points

$$\begin{aligned} \nu_1 : Y_1 &\to Y_2 \\ z &\mapsto \bar{z} \end{aligned}$$



and similarly for W_1, W_2 with a map $\nu_2 : W_1 \to W_2$.

Now to glue them together, we define

$$X_0 := \frac{U_1 \sqcup U_2 \sqcup Y_1 \sqcup Y_2 \sqcup W_1 \sqcup W_2}{z \sim \nu_1(z), y \sim \nu_2(y)}$$

(disjoint union and quotient topology).

This X_0 has a natural Riemann surface structure

- where $U_1, U_2, Y_1, Y_2, W_1, W_2$ form charts
- where there is a **holomorphic** function

  $$\rho : X_0 \to \mathbb{C}$$

  which is $\sqrt{f}, -\sqrt{f}$ glued using pasting lemma
- which is **biholomorphic** to the affine part

  $$\begin{aligned} X_0 &\to E_0 := E \cap \mathbb{C}^2 = E \setminus \{O\} \\ x &\mapsto (x, \rho(x)) \end{aligned}$$

  of the elliptic curve $E \subset \mathbb{C}P^2$

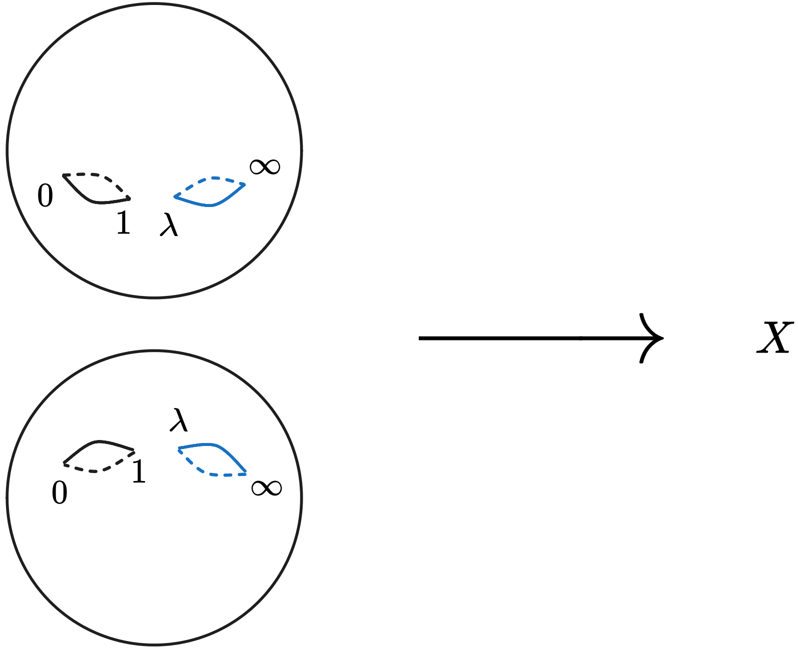
the chart at $O \in E$

The *point at infinity* $O = [0,1,0] \in E$ is not covered by any of the previous charts, so for a small enough neighborhood Y_O of O we use v in

- $$\begin{aligned} \{u_2 \neq 0\} \subseteq E &\to \mathbb{C} \\ [u_1, u_2, u_3] &\mapsto v := \frac{u_1}{u_2} \\ w &:= \frac{u_3}{u_2} \end{aligned}$$

as a chart.

We may add the point ∞ to X_0 and define X by a similar procedure



and then X will be biholomorphic to E

$$X \cong E$$

and because X is a compact surface which is a sphere with one handle-body, it must be of **genus 1**, thus a torus $S^1 \times S^1$.

Hence,

$$\pi_1(E) \cong H_1(E, \mathbb{Z}) \cong \mathbb{Z}^2$$

which we shall need in what follows.

3: holomorphic 1-form on the Elliptic curve

Definition. Holomorphic 1-forms on Riemann surfaces

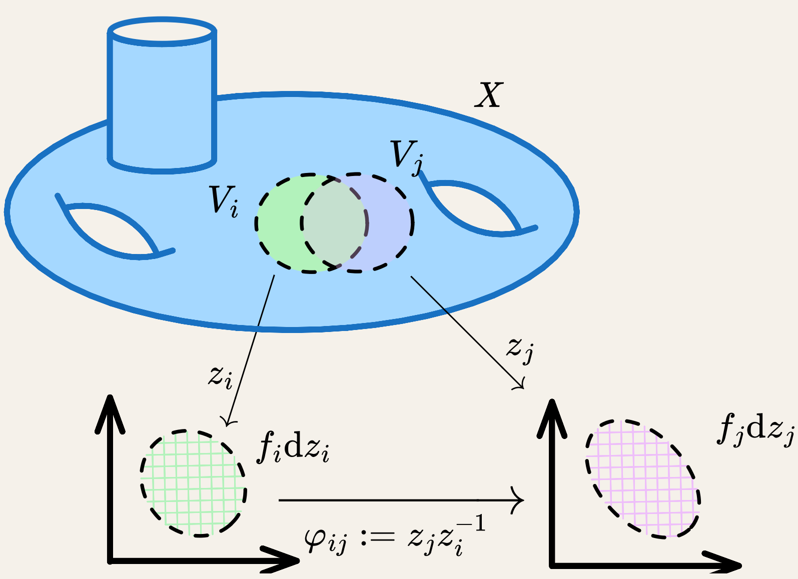
A **holomorphic 1-form** on a Riemann surface X with an atlas $\{z_i : V_i \subseteq X \to \mathbb{C} \mid i \in I\}$ is a collection of holomorphic functions

$$\{f_i : V_i \to \mathbb{C}\}$$

such that the object

$$f(z) := f_i(z)\mathrm{d}z_i \text{ for } z \in V_i$$

is *well-defined*, that is on the intersection of two charts $z_i, z_j : V_i, V_j \subseteq X \to \mathbb{C}$



we have

$$\begin{aligned}
&\text{“}f_i(z)\mathrm{d}z_i = f_j(z)\mathrm{d}z_j\text{”} \\
&\text{or “}f_i\mathrm{d}z_i = f_j\mathrm{d}(\varphi_{ij} \circ z_i)\text{”} \\
&\text{that is } f_i = \frac{\partial \varphi_{ij}}{\partial z} f_j \text{ on } V_i \cap V_j
\end{aligned}$$

- The coordinates $x, y : E \to \mathbb{C}$ are holomorphic functions on E_0 and thus by

$$\begin{aligned} y^2 - f(x) &= 0 \text{ on } E_0 \\ 2y\mathrm{d}y - \partial_x f \mathrm{d}x &= 0 \\ \frac{\mathrm{d}x}{y} &= \frac{2\mathrm{d}y}{\partial_x f} \text{ when } y \neq 0, \partial_x f \neq 0 \end{aligned}$$

  the formula "$\mathrm{d}x/y$" defines a holomorphic 1-form on E_0 on the charts given by x, y because y and $\partial_x f$ both cannot vanish together (otherwise f would have multiple roots).
  - In the local coordinates on V_1, V_2, as $y = \pm\sqrt{f(x)} \neq 0$ we have

$$\frac{\mathrm{d}x}{y} = \pm \frac{\mathrm{d}x}{\sqrt{f(x)}}$$

    for U_1, U_2 respectively.
  - In particular, the holomorphic 1-form defined by

$$\frac{\mathrm{d}x}{y}$$

    on E_0 does not vanish at any point as y and $\partial_x f$ are never zero simultaneously.
- Near the point $[0,1,0] = O \in E$, we have
  - $$\begin{aligned} \{u_2 \neq 0\} \subseteq E &\to \mathbb{C} \\ [u_1, u_2, u_3] &\mapsto v := \frac{u_1}{u_2} \\ w &:= \frac{u_3}{u_2} \end{aligned}$$

  which satisfy

$$\begin{aligned} x &= \frac{v}{w} \\ y &= \frac{1}{w} \end{aligned} \implies v = \frac{x}{y}$$

  on the intersection of their domain of definition.

  Thus

$$\frac{\mathrm{d}x}{y} = w\mathrm{d}\left(\frac{v}{w}\right) = \mathrm{d}v - \frac{1}{w}\frac{\partial w}{\partial v}\mathrm{d}v = \left(1 - \frac{1}{w}\frac{\partial w}{\partial v}\right)\mathrm{d}v$$

  - Using

$$1 = w^2 f\left(\frac{v}{w}\right) \implies w = v^3 + av^2w + bvw^2$$

    one can show

$$1 - \frac{1}{w}\frac{\partial w}{\partial v} \in \mathbb{C} \setminus \{0\}$$

> **Proposition:** Let $E \subset \mathbb{C}P^2$ be an elliptic curve (a Riemann surface), then
>
> $$\frac{\mathrm{d}x}{y}$$
>
> defines a holomorphic 1-form on E, which **vanishes nowhere** on E.

integrating the holomorphic 1-form on the Elliptic curve

Definition. Integral of a holomorphic 1-form

For a holomorphic 1-form ω, that is $\{f_i \mathrm{d}z_i\}$, on a Riemann surface X and a path $\gamma : [0,1] \to X$, we define its integral

$$\int_\gamma \omega = \sum_i \int_{z_i \circ \gamma_i} \omega_i$$

for a decomposition $\gamma \sim \gamma_1 \ldots \gamma_n$ into paths

$$\gamma_i : [0,1] \to V_i \subset X$$

Given a $\mathcal{C}^1$ path γ that takes $O \in E$ to an arbitrary element P we may integrate

$$\int_\gamma \frac{\mathrm{d}x}{y}$$

Proposition: Let $E \subset \mathbb{C}P^2$ be an elliptic curve. Then

$$\Lambda := \left\{ \int_\delta \frac{\mathrm{d}x}{y} \middle| \delta \in H_1(E, \mathbb{Z}) \right\}$$

is a rank 2 lattice in $\mathbb{C}$.

proof. Consider the two generators α_1, α_2 of $H_1(E, \mathbb{Z}) \cong \mathbb{Z}^2$.

- Suppose $\exists a \in \mathbb{R}$ such that

$$a\omega_1 = \omega_2 \iff a \int_{\alpha_1} \frac{\mathrm{d}x}{y} = \int_{\alpha_2} \frac{\mathrm{d}x}{y}$$

- Then taking conjugates

$$a \int_{\alpha_1} \overline{\frac{\mathrm{d}x}{y}} = \int_{\alpha_2} \overline{\frac{\mathrm{d}x}{y}}$$

- Thus

$$\frac{\int_{\alpha_1} \frac{\mathrm{d}x}{y}}{\int_{\alpha_1} \overline{\frac{\mathrm{d}x}{y}}} = \frac{\int_{\alpha_2} \frac{\mathrm{d}x}{y}}{\int_{\alpha_2} \overline{\frac{\mathrm{d}x}{y}}} = \frac{\omega_1}{\bar{\omega}_2} =: \lambda$$

- Then defining $\Omega := \frac{\mathrm{d}x}{y}$ and $\Phi := \bar{\lambda} \frac{\mathrm{d}x}{y}$ we have

$$\int_{\alpha_i} \Omega + \Phi = 0 \text{ for } i = 1, 2$$

which implies $\Omega + \Phi$ is exact, thus by

PROPOSITION 2.6. *Suppose C is a compact Riemann surface, and $\omega, \varphi \in \Omega^1(C)$. Considering ω and φ as differential one-forms on C, if*

$$\omega + \bar{\varphi} = df,$$

where f is a C^∞ function on C, then

$$\omega = \varphi = 0.$$

implies

$$\frac{\mathrm{d}x}{y} \equiv 0$$

which is a contradiction.

- *The proof of the above proposition is as follows:* Say $\Omega = h(z)\mathrm{d}z$ and $\Phi = g(z)\mathrm{d}z$ then

$$\Omega + \bar{\Phi} = \mathrm{d}q$$

and

$$\Omega \wedge \Phi = h(z)g(z)\mathrm{d}z \wedge \mathrm{d}z$$

  - So

$$\begin{aligned} \Phi \wedge \bar{\Phi} &= |g(z)|^2 \mathrm{d}z \wedge \mathrm{d}\bar{z} \\ \text{also} &= \Phi \wedge (\Omega - \mathrm{d}q) = \Phi \wedge \mathrm{d}q \end{aligned}$$

  - and

$$\mathrm{d}(q\Phi) = \mathrm{d}q \wedge \Phi + f\,\cancelto{0}{\mathrm{d}\Phi}$$

  - This means

$$|g(z)|^2 \mathrm{d}z \wedge \mathrm{d}\bar{z} = -\mathrm{d}(q\Phi)$$

    but that would imply

$$\int_M |g(z)|^2 \mathrm{d}z \wedge \mathrm{d}\bar{z} = 0 \implies g = 0 \implies \bar{\Phi} = 0$$

    as $\frac{i}{2}\mathrm{d}z \wedge \mathrm{d}\bar{z}$ is a real 2-form.

- Thus ω_1, ω_2 are $\mathbb{R}$-linearly independent.

4: the period map

> **Proposition:** The period map
>
> $$\int \frac{\mathrm{d}x}{y} : E \to \frac{\mathbb{C}}{\Lambda_\omega}$$
>
> is a well-defined **holomorphic** map between Riemann surfaces whose derivative **vanishes nowhere** on E.

Because

> **Proposition:** Let $E \subset \mathbb{C}P^2$ be an elliptic curve (a Riemann surface), then
>
> $$\frac{\mathrm{d}x}{y}$$
>
> defines a holomorphic 1-form on E, which **vanishes nowhere** on E.

on every chart V, W on E, $\frac{\mathbb{C}}{\Lambda}$ respectively we have a $\mathcal{C}^1$

$$h : V \to W$$

such that

$$\frac{\mathrm{d}x}{y} = g(z)\mathrm{d}z$$

with $g \neq 0$ holomorphic on V and

$$h'(z) = g(z) \neq 0$$

- Thus

$$h : V \to W$$

is holomorphic $\implies$ the map $E \to \mathbb{C}/\Lambda_\omega$ is holomorphic.

> **Proposition:** The period map
>
> $$\int \frac{\mathrm{d}x}{y} : E \to \frac{\mathbb{C}}{\Lambda_\omega}$$
>
> is a biholomorphism of **compact** Riemann surfaces.

By

> **Proposition:** The period map
>
> $$\int \frac{\mathrm{d}x}{y} : E \to \frac{\mathbb{C}}{\Lambda_\omega}$$
>
> is a well-defined **holomorphic** map between Riemann surfaces whose derivative **vanishes nowhere** on E.

we conclude by **compactness of E** that the map is a topological covering map of the torus $\mathbb{C}/\Lambda_\omega$.

- This means there are maps

$$\begin{array}{ccc} \mathbb{C} & & \\ \downarrow & \searrow & \\ E & \to & \mathbb{C}/\Lambda_\omega \end{array}$$

where $\mathbb{C} \to E$ is quotient by some (rank 1 or 2) lattice $L \subset \mathbb{C}$.

- Now the induced map on the fundamental groups

$$\pi_1(E, O) \hookrightarrow \pi_1(\mathbb{C}/\Lambda_\omega, 0)$$

lifts to the lattices

$$\begin{array}{ccc} \pi_1(E, O) & \hookrightarrow & \pi_1(\mathbb{C}/\Lambda_\omega, 0) \\ \Big\downarrow{\scriptstyle \gamma \mapsto \tilde{\gamma}(1)} & & \Big\downarrow{\scriptstyle \gamma \mapsto \tilde{\gamma}(1)} \\ L & \hookrightarrow & \Lambda_\omega \subset \mathbb{C} \end{array}$$

where $\tilde{\gamma}$ is the unique lift of γ to $\mathbb{C}$

- But

$$\tilde{\gamma}(1) = \int_\gamma \omega$$

and

$$L = \{\tilde{\gamma}(1) \mid \gamma \in \pi_1(\mathbb{C}/\Lambda_\omega, 0)\} = \left\{ \int_\gamma \omega \mid \gamma \in \pi_1(E, O) \right\} = \Lambda_\omega$$

- Thus

$$\pi_1(E, O) \hookrightarrow \pi_1(\mathbb{C}/\Lambda_\omega, 0)$$

is **surjective**.

- Because topological coverings $X \to \mathbb{C}/\Lambda_\omega$ are classified by the image of $\pi_1(X)$ in $\pi_1(\mathbb{C}/\Lambda_\omega)$, we conclude E is the trivial covering of $\mathbb{C}/\Lambda_\omega$ thus

$$\int \frac{\mathrm{d}x}{y} : E \cong \frac{\mathbb{C}}{\Lambda_\omega}$$

Simon Donaldson #book on Riemann surfaces

Chapter 7

Proposition: Let

$$X := V(p) \subseteq \mathbb{C}P^2$$

be a smooth complex curve of degree d for the homogeneous polynomial p with non-vanishing (on X) partial derivatives. Then the genus of X is

$$g_X = \frac{1}{2}(d-1)(d-2)$$

So this means

degree of $p(X) \in \mathbb{C}[X]$		$g_X(d)$	Riemann surface $V(p)$
1		0	sphere
2	conic	0	sphere
3	cubic	1	torus
nope!		2	
4		3	

As

$$p(x,y) = 0$$

on X we have

$$p_x \mathrm{d}x + p_y \mathrm{d}y = 0$$
$$\theta := \frac{\mathrm{d}x}{p_y} = -\frac{\mathrm{d}y}{p_x}$$

as p_x, p_y do not vanish on X.

Chapter 11

algebraic degree and degree of meromorphic maps

Let Σ be a compact Riemann surface. It admits a non-trivial holomorphic

$$f : \Sigma \to \mathbb{C}P^1$$

Then

$$\begin{aligned} f_* : \mathbb{C}[\mathbb{C}P^1] \cong \mathbb{C}(z) &\hookrightarrow \mathbb{C}(\Sigma) \\ p(z) &\mapsto p(f(-)) \end{aligned}$$

is an embedding of fields.

$$[\mathbb{C}(\Sigma), \mathbb{C}(z)] = \deg f$$

Let (by translation if necessary) $0 \in \mathbb{C}P^1$ is a regular value of f so $f^{-1}(0)$ is a set of $d := \deg f$ distinct points

$$p_1, \ldots, p_d \in \Sigma$$

- By *existence* for each i there is a meromorphic

  $$g_i : \Sigma \to \mathbb{C}P^1$$

  that has a simple pole at p_i but is holomorphic around p_j for each $j \neq i$.
- Suppose

  $$\sum_j \lambda_j g_j = 0$$

  where $\lambda_j \in \mathbb{C}(z) \hookrightarrow \mathbb{C}(\Sigma)$. By multiplying a suitable z^k we can assume λ_j are holomorphic around $z = 0$ and do not all vanish at $z = 0$.
- Consider

  $$\sum_j \lambda_j(f(p_i))g_j(p_i) = 0$$

  The only non-holomorphic term is $\lambda_i g_i$ so

  $$\lambda_j(\underbrace{f(p_i)}_{0}) = 0$$

  which is a contradiction.
- Thus

  $$[\mathbb{C}(\Sigma), \mathbb{C}(z)] \geq \deg f$$

- By *primitive element theorem* it suffices to show any $g \in \mathbb{C}(\Sigma)$ satisfies

  $$P(f, g) = \lambda_0(f) + g\lambda_1(f) + g^2\lambda_2(f) + \cdots + g^d\lambda_d(f)$$

  - Let z be a regular value of f so

    $$f^{-1}(z) = \{p_1, \ldots, p_d\} \subset \Sigma$$

    such that

    $$w_i := g(p_i) \in \mathbb{C} \subset \mathbb{C}P^1$$

  - If

    $$a_1 := \sum_i w_i$$

    and so on are elementary symmetric functions we have

    $$\sum_{r=0}^{d} (-1)^r a_r g^r = 0$$

holomorphic maps and field extensions

Corollary of

Let Σ be a compact Riemann surface. It admits a non-trivial holomorphic

$$f : \Sigma \to \mathbb{C}P^1$$

Then

$$\begin{aligned} f_* : \mathbb{C}[\mathbb{C}P^1] \cong \mathbb{C}(z) &\hookrightarrow \mathbb{C}(\Sigma) \\ p(z) &\mapsto p(f(-)) \end{aligned}$$

is an embedding of fields.

$$[\mathbb{C}(\Sigma), \mathbb{C}(z)] = \deg f$$

If

$$f : \Sigma_1 \to \Sigma_2$$

is a non-constant holomorphic map between compact connected Riemann surfaces then

$$[\mathbb{C}(\Sigma_1), \mathbb{C}(\Sigma_2)] = \deg f$$

Consider

$$f_0 : \Sigma_2 \to \mathbb{C}P^1$$

then

$$\mathbb{C}(z) \hookrightarrow \mathbb{C}(\Sigma_2) \hookrightarrow \mathbb{C}(\Sigma_1)$$

which gives

$$[\mathbb{C}(\Sigma_1), \mathbb{C}(z)] = [\mathbb{C}(\Sigma_1), \mathbb{C}(\Sigma_2)][\mathbb{C}(\Sigma_2), \mathbb{C}(z)]$$

fields of transcendence degree $= 1$

- We know

  $$\mathbb{C}(z) \hookrightarrow \mathbb{C}(\Sigma)$$

  Thus the transcendence degree is at least 1.
- Suppose

  $$\mathbb{C}(z_1, z_2) \hookrightarrow \mathbb{C}(\Sigma)$$

  then

  $$z_1 \mapsto f : \Sigma \to \mathbb{C}P^1$$

  (non-constant). By

  Let Σ be a compact Riemann surface. It admits a non-trivial holomorphic

  $$f : \Sigma \to \mathbb{C}P^1$$

  Then

  $$\begin{aligned} f_* : \mathbb{C}[\mathbb{C}P^1] \cong \mathbb{C}(z) &\hookrightarrow \mathbb{C}(\Sigma) \\ p(z) &\mapsto p(f(-)) \end{aligned}$$

  is an embedding of fields.

  $$[\mathbb{C}(\Sigma), \mathbb{C}(z)] = \deg f$$

  we know

  $$[\mathbb{C}(\Sigma), \mathbb{C}(z_1)] < \infty \implies [\mathbb{C}(z_1, z_2), \mathbb{C}(z_1)] < \infty$$

  which is a contradiction.

Let K is any field extension of $\mathbb{C}$ and of transcendence degree 1 then for every

$$\iota : \mathbb{C}(z) \hookrightarrow K$$

there is a compact connected Riemann surface Σ with $K = \mathbb{C}(\Sigma)$ and

$$f : \Sigma \to \mathbb{C}P^1$$

such that $f_* = \iota$.

valuations

$$\begin{array}{r} \text{cpt conn Riemann surfaces} \\ \text{non-constant holomorphic maps} \end{array} \leftrightarrow \begin{array}{l} \text{fields of transcendence degree 1} \\ \text{field inclusions} \end{array}$$